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Efficacy of education on injection technique for diabetes patients with lipohypertrophy: a systematic review and meta-analysis

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Efficacy of education on injection technique for diabetes patients with lipohypertrophy: a systematic review and meta-analysis

Masahiro Ichikawa[a]*, Tomoaki Akiyama[a], Yasushi Tsujimoto[b], Keisuke Anan[c], Tadashi Yamakawa[a], Yasuo Terauchi[d]

[a]Department of Endocrinology and Diabetes, Yokohama City University Medical Center, Yokohama, Japan

[b]Department of Nephrology and Dialysis, Kyoritsu Hospital, Kawanishi, Japan

[c]Department of Healthcare Epidemiology, Graduate School of Medicine and Public Health, Kyoto University, Kyoto, Japan

[d]Department of Endocrinology and Metabolism, Yokohama City University School of Medicine, Yokohama, Japan

Email Addresses:

***Corresponding Author**

Masahiro Ichikawa

Department of Endocrinology and Diabetes

Yokohama City University Medical Center

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Urafunecho 4-57, Minami-ku

Yokohama 232-0024, Japan

Tel: +81-45-261-5656

Fax: +81-45-231-1846

E-mail: t196009c@yokohama-cu.ac.jp

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ABSTRACT

Objectives: This study aimed to investigate the efficacy of providing education on injection technique to diabetes patients with lipohypertrophy (LH).

Design: Systematic review and meta-analysis

Methods: We included diabetes patients who use insulin and have LH, and excluded patients without LH. We performed a literature search on CENTRAL, MEDLINE, EMBASE, ICTRP, and ClinicalTrials.gov on March 2021 for randomised controlled trials (RCTs). We used the revised Cochrane Risk of Bias 2 tool to evaluate the risk of bias in each outcome in each study. We pooled the data using a random-effects model and evaluated the certainty of evidence using the Grading of Recommendations Assessment, Development, and Evaluation approach.

Outcome measures: The primary endpoints were change in total daily dose (TDD) of insulin, change in HbA1c levels, and prevalence of hypoglycaemia.

Results: We screened 707 records and included two RCTs (319 participants) in the meta-analysis. Education on injection technique may slightly increase the change of TDD of insulin (2 studies, 319 participants: mean difference (MD) -4.53; 95% confidence interval (CI) -8.5, -0.57; $p = 0.03$; $I^2 = 0\%$; low certainty of evidence) and may not result in an increased change in HbA1c compared to that in the control group (two studies, 319

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participants: MD -0.12; 95% CI -0.36, 0.12; p = 0.32; I^2 = 0%; low certainty of evidence). Providing education about injection technique may have little-to-no effect on the prevalence of hypoglycaemia (two studies, 319 participants: risk ratio 0.99; 95% CI 0.27, 3.70; p = 0.99; I^2 = 29%; very low certainty of evidence].

Conclusions: The present meta-analysis suggests that injection technique education may result in a slight reduction in the TDD of insulin without increasing HbA1c, and clinicians could provide education on injection technique for diabetes patients with LH.

Protocol registration: DOI: dx.doi.org/10.17504/protocols.iobtiinkce

Keywords: injection technique, lipohypertrophy, diabetes mellitus, education

Strengths and limitations of this study

- To the best of our knowledge, this is the first systematic review and meta-analysis to investigate the effect of injection technique education in diabetes patients with lipohypertrophy.
- We conducted a comprehensive search and summarised the evidence for the effects of injection technique education, used a prospective registered protocol, followed the Cochrane Handbook and GRADE, selected a rigorous methodology, and adhered to reporting guidelines.
- The limitation of this study is that we did not have sufficient data to perform a subgroup

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analysis.

INTRODUCTION

There has been a continuous increase in the number of diabetes patients worldwide, and diabetes has become a major public health problem, with 4.2 million adults estimated to die due to diabetes and its complications.[1] The number of diabetes patients is expected to increase to 700 million by 2045.[1] Blood glucose control is important to prevent the complications of diabetes, and insulin therapy plays a key role in regulating blood glucose level in patients.[2] Insulin therapy is necessary not only for type 1 diabetes but also for type 2 diabetes and is estimated to be used in 7.5% of type 2 diabetes patients.[3] Lipohypertrophy (LH), a side effect of long-term insulin injection, has been reported in 38% of diabetes patients.[4] LH is caused by the thickening of adipose tissue localised to the site of insulin injection.[5, 6]

Patients with LH face difficulty with this condition due to its impact on insulin variability.[4, 7] Both insulin absorption and action becomes slow, and insulin variability greatly increases between injections in those with LH;[8] moreover, the risk of unexpected hypoglycaemia has been noted in patients with LH and is reported to be approximately six times higher than that in patients without LH.[7] Previous studies have

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suggested that providing proper education on injection techniques may reduce HbA1c levels and/or total daily dose (TDD) of insulin.[9-11] Injection technique education includes rotating the injection sites properly, avoiding needle reuse, and avoiding injecting into the site of LH.[5] It is also recommended to use a 4-mm-long needle to avoid injection into the muscle.[5, 12]

Two RCTs were conducted in diabetes patients with LH who received education on injection technique.[13, 14] To the best of our knowledge, there has been no systematic investigation on the effects of providing education about injection technique on the TDD of insulin, HbA1c levels, and hypoglycaemia in diabetes patients with LH. Therefore, we aimed to perform a systematic review and meta-analysis to determine the efficacy of providing education on the injection technique to diabetes patients with LH.

METHODS

Ethics approval

As this was a systematic review, there were no ethical issues involved. We registered the protocol in protocols.io (DOI: dx.doi.org/10.17504/protocols.io.btiinkce). Patient consent was not necessary for the present study.

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Eligibility criteria

We conducted this systematic review in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines,[15] and the checklist is shown in the online supplemental table S1. We conducted a systematic review investigating the effects of injection technique education on patients with type 1 or type 2 diabetes who use insulin and have LH, and included individual, cluster, and crossover randomised trials. We did not limit the inclusion to languages or countries. We included published and unpublished papers, conference abstracts, letters, and all other types of papers. We excluded observational studies and non-randomised controlled trials (RCTs). We did not exclude studies because of the observation period or year of publication.

We included patients with type 1 or type 2 diabetes who had been receiving insulin injections for at least 1 year, with LH, 18 years old and above, any sex, no HbA1c restriction, and no education about LH within the past 6 months. LH diagnosis was made by a physician, nurse, or investigators when it was clinically visible, palpable, or visible on ultrasound.[4] We accepted any number of insulin injections per day or needle size for insulin injections.

We excluded pregnant women, patients who wished to become pregnant, lactating women, and patients taking medications that may cause LH (anti-retroviral or

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corticosteroid therapy).

Information sources

Search strategy

We searched the Cochrane Central Register of Controlled Trials (CENTRAL) on 17 March 2021, and MEDLINE through PubMed and EMBASE by ProQuest Dialog on 16 March 2021. We also searched the World Health Organization International Clinical Trials Platform Search Portal (ICTRP) on 18 March 2021 and ClinicalTrials.gov on 16 March 2021 for ongoing or unpublished trials. The search strategies are presented in the online supplemental Table S2. We checked the reference lists of the studies, including international guidelines, and identified the reference lists of eligible studies and articles citing eligible studies.

Selection process

Two independent reviewers (MI and TA) screened the titles and abstracts, followed by an assessment of eligibility based on the full texts. We contacted the original authors if the relevant data were missing. Disagreements between the two reviewers were resolved by discussion, and if this was not possible, a third reviewer (YT or KA) became an arbiter.

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Data collection process

Two reviewers (MI and TA) independently extracted data from included studies using a standardised data collection form. Disagreements were resolved by discussion, and if this was not possible, a third reviewer (YT or KA) became an arbiter. We conducted up to two author queries by e-mail to obtain relevant data from the researchers.

Data items

The primary endpoints were changes in insulin TDD and HbA1c levels, and the prevalence of hypoglycaemia. The change in insulin TDD was defined as the change in TDD from baseline, and the duration was the longest follow-up period after 3 months. The change in HbA1c levels was defined as the change in HbA1c from baseline, and the duration was the longest follow-up period after 3 months. Hypoglycaemia was defined as one or more hypoglycaemic symptoms (palpitations, fatigue, sweating, strong hunger, dizziness, tremors, etc.) and a blood glucose level below 60 mg/dL.[7] We accepted hypoglycaemia as defined by the original authors. Duration was defined as the follow-up period. The secondary endpoints were changes in the proportion of patients with cured LH and all adverse events. The definition of change in the proportion of patients with

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cured LH was the same as that of the original paper, and the duration was the longest follow-up period after 3 months. All adverse events were the same as those defined in the original study.

The data collecting form included information on the first author's name, year of publication, country, sample size, proportion of male participants, mean age of participants, number of participants with type 1 diabetes mellitus (T1DM)/type 2 diabetes mellitus (T2DM), estimated diabetes duration, mean duration of insulin treatment, intervention details, detection methods of LH, and the outcome measures mentioned above. We asked the authors of the original papers for unpublished or additional data and obtained research data for Chen 2021 but not for Campinos 2017.

Study risk of bias assessment

Two reviewers (MI and TA) independently assessed the risk of bias using the Risk of Bias 2 (RoB2) tool.[16] The effect of interest is the intention-to-treat effect, which is the effect assigned to the interventions at baseline, regardless of whether or not the intervention was received as intended. Disagreements between the two reviewers were discussed, and if this was not possible, a third reviewer (YT or KA) became an arbiter.

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Effect measures

We pooled the relative risk ratios and 95% confidence intervals (CIs) for the following binary variables: hypoglycaemia, proportion of patients with cured LH, and all adverse events. We pooled the mean differences and 95% CIs for the following continuous variables: HbA1c and TDD of insulin. Although we did not conduct a meta-analysis, we summarised adverse events according to the definition in the original paper.

We extracted the data on an intention-to-treat basis for all dichotomous data whenever possible. For continuous data, we did not impute missing data based on the recommendation of the Cochrane Handbook.[17] We performed a meta-analysis of the data available in the original study.

When the original studies reported only standard errors or p-values, we used Altman's method to calculate the standard deviation.[18] When these values were not available after enquiring the authors, we calculated the standard deviations using confidence intervals and t-values based on the methods in the Cochrane Handbook [17] or validated methods.[18] We analyzed the validity of these methods using sensitivity analysis.

Statistical analysis

We used Review Manager software (RevMan 5.4; The Nordic Cochrane Centre, The

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Cochrane Collaboration, Copenhagen, Denmark) to conduct a meta-analysis using the DerSimonian and Laird random-effects method. We assessed statistical heterogeneity by visual inspection of the forest plots and calculation of I^2 statistic (I^2 values of 0-40%: might not be important; 30-60%: may represent moderate heterogeneity; 50-90%: may represent substantial heterogeneity; 75-100%, considerable heterogeneity). For the I^2 statistic, we performed the Cochrane Chi^2 test (Q test) and considered it statistically significant if the P value was less than 0.10.

Certainty assessment

Based on the Cochrane Handbook,[17] we created a summary of findings (SoF) table for the results of changes in the insulin TDD, changes in HbA1c levels, and the prevalence of hypoglycaemia. A SoF table contained important information on the relative and absolute effect sizes of the interventions investigated, the amount of available evidence, and the certainty of available evidence.[17]

We assessed the quality of evidence for each outcome based on the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) method.[19] GRADE specifies four categories, namely high, moderate, low, and very low, which are applied to accumulating evidence. Evidence from randomised trials starts with high

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certainty of evidence and can be downgraded by five domains: risk of bias, imprecision, inconsistency, indirectness, and publication bias.

Difference between a protocol and a review

To clarify the influence of effect modifiers on the results, subgroup analyses of the primary outcome were planned to be evaluated with the following factors when sufficient data were available: age (< 65 years vs. ≥ 65 years), baseline HbA1c level of < 8% vs. ≥ 8%,[20] sex, and injection technique education protocol (IT) with or without initial insulin level reduction.

We also planned to conduct the following sensitivity analyses on the primary outcomes to assess whether the results of the review were robust to the decisions made during the review process: exclusion of studies with imputed statistics, and exclusion of studies with an overall high risk of bias. We could not perform a predetermined subgroup analysis because sufficient data were not available. We also could not perform a predetermined sensitivity analysis because neither study used imputed statistics and had a high overall risk of bias.

We planned to assess the possibility of publication bias by visually inspecting the funnel plots and testing the Egger's test. If there were fewer than 10 trials or trials with similar

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sample sizes, then we decided not to perform the test. We could not perform the Egger test for reporting bias detection because there were fewer than 10 trials.

Patients and public involvement

As this was a systematic review, patients or the public were not involved in our research.

RESULTS

Study selection

We performed the search in March 2021 and found 707 records; after duplicates were removed, 657 records were screened for titles and abstracts. Seven full-text records were assessed for eligibility. Five studies were excluded, and the reasons were as follows: four for including the wrong population and one for using the wrong design. Details of the excluded studies are shown in the online supplemental Table S3. Finally, we included two RCTs that met the eligibility criteria (Figure 1).

Study characteristics

Table 1 presents the characteristics of the two included studies: one conducted in France and the other in China. The sample size was 123 cases in France and 210 cases in China.

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The detection method of LH was visible and/or palpable LH as determined by nurses in the French study, and ultrasound was added to the French method in the study conducted in China. Mean patient age was 52.8 ± 15.1 years in the IT group and 51.4 ± 16.4 years in the control (C) group for the French study, and 59 ± 7 years in the IT group and 60 ± 9 years in the C group for the Chinese study. In the French study, males accounted for 70.7% of the total sample, and the number of T1DM/T2DM individuals was 66/57, whereas in the Chinese study, males accounted for 46.7% of the total sample, and the number of T1DM/T2DM individuals was 7/203. DM duration and insulin treatment in the French study were 19.8 ± 11.5 years and 14.8 ± 12.8 years vs. 16.3 ± 9.2 years and 14.8 ± 12.8 years in the IT and C groups, respectively. The DM duration and insulin treatment in the Chinese study were 14.4 ± 5.8 years and 6.9 ± 4.7 years vs. 16.8 ± 7.6 years and 7.1 ± 5.4 years in the IT and C groups, respectively. The interventions were the same in both France and China studies.

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Table 1 Characteristics of the included studies

Study	Methods	Subject characteristics	Intervention details	Reported outcome of interest
Campions 2017	Location: France Sample size: 123 Detection method of LH: visible and/or palpable LH determined by nurses	Age (years): 52.8 ± 15.1 (IT), 51.4 ± 16.4 (C) Male (%): 70.7 Number of T1DM/T2DM individuals: 66/57 Estimated diabetes duration (years): 19.8 ± 11.5 (IT), 16.3 ± 9.2 (C) Insulin treatment duration (years): 14.8 ± 12.8 (IT), 12.5 ± 10 (C)	IT: reducing insulin doses initially by 20%, instructing to rotate within injection sites, foregoing needle reuse, and switching to 4-mm needles. C: standard care	Change in total daily doses of insulin from baseline Change in HbA1c levels from baseline Hypoglycaemia Change in the proportion of patients with cured LH All adverse events
Chen 2021	Location: China Sample size: 210 Detection method of LH: visible and/or palpable LH determined by nurses and ultrasound examination	Age (years): 59 ± 7 (IT), 60 ± 9 (C) Male (%): 46.7 Number of T1DM/T2DM individuals: 7/203 Estimated diabetes duration (years): 14.4 ± 5.8 (IT), 16.8 ± 7.6 (C) Insulin treatment duration (years): 6.9 ± 4.7 (IT), 7.1 ± 5.4 (C)	IT: reducing insulin doses initially by 20%, instructing to rotate within injection sites, foregoing needle reuse, and switching to 4-mm needles. C: standard care	Change in the total daily doses of insulin from baseline Change in HbA1c levels from baseline Hypoglycaemia Change in the proportion of patients with cured LH All adverse events

DM, Diabetes mellitus; IT, Injection technique education group; C, Control group; LH, lipohypertrophy

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Risk of bias in the studies

Figure 2 shows the RoB2 result: In both RCTs, domain 2 (bias due to deviations from the intended intervention) and/or domain 3 (bias due to missing outcome data) were at high risk, and the overall risk of bias was high.

Outcome and certainty of evidence

Table 2 shows a summary of these findings. Figure 3 shows a forest plot. Compared with standard care, injection technique education slightly increased the change in TDD of insulin (two studies, 319 participants: MD -4.53; 95% CI -8.5, -0.57; p = 0.03; I^2 = 0%; low certainty of evidence). The certainty of evidence was low because of the high risk of bias, not meeting the optimal information size criterion. Compared with standard care, injection technique education may not increase the change in HbA1c levels (two studies, 319 participants: MD -0.12; 95% CI -0.36, 0.12; p = 0.32; I^2 = 0%; low certainty of evidence). The quality of evidence was low because of the serious risk of bias and imprecision.

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Table 2 Summary of findings comparing indicated outcomes between injection technique education and standard care

Patient or population: Patients with type 1 diabetes or type 2 diabetes who use insulin and have LH

Setting: Outpatient

Intervention: Injection technique education

Comparison: Standard care

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Quality of evidence (GRADE)
	Assumed risk-C	Corresponding risk-IT			
Change in TDD of insulin	Mean change in the TDD of insulin in the C group was -3.12 to -0.9.	Mean change in the TDD of insulin in the IT group was 4.53 lower (8.50 lower to 0.57 lower)		319 (2)	Low[a,b]
Change in HbA1c level	Mean change in HbA1c level in the C group was -0.55 to -0.54.	Mean change in HbA1c level in the IT group was 0.12 lower (0.36 lower to 0.12 higher)		319 (2)	Low[a,b]
Prevalence of hypoglycaemia	9/160 (5.63%)	8/159 (5.03%)	RR 0.99 (0.27 to 3.70)	319 (2)	Very low[a,c]

* Corresponding risk (and 95% CI) are based on assumed risks in the control group and relative risks (and 95% CI) of the intervention.
[a] Due to high risk of bias
[b] Due to not meet the optimal size criterion
[c] Due to not meet the optimal size criterion and 95% CI of less than 0.75 and greater than 1.25
LH, lipohypertrophy; C, Control; IT, Injection technique education; TDD, total daily doses; HbA1c, haemoglobin A1c; RR, risk ratio; CI, confidence interval. GRADE Working Group grades of evidence: High quality: we are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect. Very low quality: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

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The evidence is very uncertain about the effect of insulin injection technique on the prevalence of hypoglycaemia (two studies, 319 participants: risk ratio [RR] 0.99; 95% CI 0.27, 3.70; p = 0.99; I^2 = 29%; very low certainty of evidence), and the prevalence of cured LH compared to the control group (two studies, 319 participants: RR 1.19; 95% CI 0.73, 1.92; p = 0.49; I^2 = 0%; very low certainty of evidence). The very low certainty of evidence was due to the seriously high risk of bias and serious imprecision.

DISCUSSION

The present study showed that injection technique education may result in a slight reduction in the TDD of insulin. Injection technique education may not increase the change in HbA1c levels, and the evidence is very uncertain about the effect of insulin injection technique on the prevalence of hypoglycaemia and the prevalence of cured LH compared to those in the control group. To the best of our knowledge, this is the first systematic review and meta-analysis to study the effects of insulin injection technique education in comparison with standard care for diabetes patients with LH.

The results of the present study were partly consistent with those of previous observational studies, which found that injection technique education was associated with changes in TDD of insulin and HbA1c levels.[9, 10] These previous observational studies reported that injection technique education decreased the TDD of insulin and HbA1c levels in patients with diabetes upon insulin injection. Our meta-analysis of randomised trials showed similar results. Meanwhile, the results of the present study did not show a decrease in HbA1c levels, but this might be due to the fact that the two studies included were initially conducted with up to 20% reduction in TDD of insulin to prevent hypoglycaemia.

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Excessive exogenous insulin can be harmful, and it is important to reduce the TDD of insulin to prevent adverse events. Excessive exogenous insulin administration has been reported to be associated with adverse events such as hypoglycaemia, weight gain, and increased cardiovascular events and mortality.[21, 22] A meta-analysis in type 1 diabetes patients integrated aerobic-only intervention studies in adults, and multiple types of exercise interventions, including aerobic and anaerobic, in 12- to 18-year-olds. The results showed a reduction of 0.21 U/kg in TDD of insulin.[23] In the present study, the TDD of insulin was reduced by 4.53 U/day, which is less than that obtained with exercise intervention, which is effective in avoiding excessive insulin administration.

The certainty of evidence in the present study was low or very low. In both included studies, a portion of the control group was shifted to the ideal injection method, which contributed to the effect of high risk of bias due to deviation from the intended intervention. However, even if some of those from the control group who were shifted to the ideal injection technique group, the present study showed a slight reduction in the TDD of insulin, which underestimates the effect; in fact, the effect may be even greater. In both studies, the bias due to missing outcome data for the continuous variable outcome was high because the missing value was greater than 5%. The bias due to missing outcome data for the binary variable outcome was low or high for the frequency of outcome and number of missing outcomes. The registration of the prevalence of hypoglycaemia in clinicaltrials.gov mentions only a timeframe of 6 months, but because the prevalence of hypoglycaemia was counted at 3 and 6 months, we considered the bias in the selection of the results reported by Campinos 2017 as high. The other reason for lowering the GRADE is imprecision, because only two studies were included, and the sample size was small. In the future, more studies with sufficient sample sizes are needed.

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The strengths of this study are that we conducted a comprehensive search and summarised the evidence for the effects of injection technique education, used a prospective registered protocol, followed the Cochrane Handbook and GRADE, selected a rigorous methodology, and adhered to reporting guidelines.

However, this study had several limitations. First, as described above, we could not have enough data to conduct subgroup analyses and assess publication bias. Because our comprehensive search did not show any ongoing or unpublished studies, it was not plausible for publication bias to affect our findings. Second, the generalisability of our findings are limited. For example, the included studies tended to recruit middle-aged, obese, and type 2 diabetes patients, and measured the outcomes at 6 months. However, a crossover study limited to type 1 diabetes individuals has shown that injections administered at sites away from the LH improved insulin absorption and reduce glycaemic variability compared to injections administered at the LH.[8] Previous studies have also shown the efficacy of optimised injection technique on TDD of insulin, although baseline characteristics varied, suggesting that it may be effective in the aforementioned populations.[9, 10] Further studies evaluating different populations and long-term outcomes are required.

CONCLUSION

The present meta-analysis suggests that providing education on injection technique to diabetes patients with LH may slightly reduce the TDD of insulin. Our findings suggest that doctors may consider providing education on injection technique to diabetes patients with LH. In the future, we hope that RCTs with less bias, especially for the deviation of the intended protocol and with a rigorous protocol, will be conducted to accurately

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evaluate the effects.

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COMPETING INTERESTS:

All authors have no conflicts of interest to declare.

AUTHORS' CONTRIBUTIONS

MI conducted the search and analysed the data. MI and TA conducted screening of titles and abstracts, extracted data, and evaluated the risk of bias. If the disagreement could not be resolved, then YT or KA became an arbiter. MI, YT, and KA drafted the manuscript. YT, KA, TY, and YT revised the manuscript. All authors approved the manuscript version submitted for publication and agreed to be responsible for all aspects of this work.

ETHICAL APPROVAL

Not required.

PATIENT CONSENT FORM

Not required

DATA SHARING STATEMENT

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The dataset is available from the corresponding author upon request.

REFERENCES

1 Federation ID. IDF diabetes atlas. 9th ed. http://www.diabetesatlas.org/. (accessed 11 June 2021).

2 Brown A, Guess N, Dornhorst A, et al. Insulin-associated weight gain in obese type 2 diabetes mellitus patients: What can be done? *Diabetes Obes Metab* 2017;19:1655–68.

3 Basu S, Yudkin JS, Kehlenbrink S, et al. Estimation of global insulin use for type 2 diabetes, 2018–30: a microsimulation analysis. *Lancet Diabetes Endocrinol* 2019;7:25–33.

4 Deng N, Zhang X, Zhao F, et al. Prevalence of lipohypertrophy in insulin-treated diabetes patients: A systematic review and meta-analysis. *J Diabetes Investig* 2018;9:536–43.

5 Frid AH, Kreugel G, Grassi G, et al. New insulin delivery recommendations. *Mayo Clin Proc* 2016;91:1231–55.

6 Gentile S, Guarino G, Corte TD, et al. Insulin-induced skin lipohypertrophy in type 2 diabetes: a multicenter regional survey in Southern Italy. *Diabetes Ther* 2020;11:2001–17.

7 Blanco M, Hernández MT, Strauss KW, et al. Prevalence and risk factors of lipohypertrophy in insulin-injecting patients with diabetes. *Diabetes Metab* 2013;39:445–53.

8 Famulla S, Hövelmann U, Fischer A, et al. Insulin injection into lipohypertrophic tissue: blunted and more variable insulin absorption and action and impaired postprandial glucose control. *Diabetes Care* 2016;39:1486–92.

For peer review only

9 Grassi G, Scuntero P, Trepiccioni R, et al. Optimizing insulin injection technique and its effect on blood glucose control. *J Clin Transl Endocrinol* 2014;1:145–50.

10 Smith M, Clapham L, Strauss K. UK lipohypertrophy interventional study. *Diabetes Res Clin Pract* 2017;126:248–53.

11 Misnikova IV, Gubkina VA, Lakeeva TS, et al. A randomized controlled trial to assess the impact of proper insulin injection technique training on glycemic control. *Diabetes Ther* 2017;8:1309–18.

12 Frid AH, Hirsch LJ, Menchior AR, et al. Worldwide injection technique questionnaire study: Injecting complications and the role of the professional. *Mayo Clin Proc* 2016;91:1224–30.

13 Campinos C, Le Floch JP, Petit C, et al. An effective intervention for diabetic lipohypertrophy: Results of a randomized, controlled, prospective multicenter study in France. *Diabetes Technol Ther* 2017;19:623–32.

14 Chen L, Xing Q, Li J, et al. Injection technique education in patients with diabetes injecting insulin into areas of lipohypertrophy: A randomized controlled trial. *Diabetes Ther* 2021. doi: 10.1007/s13300-021-01013-1 [published Online First: 2021/02/12]

15 Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021:n71.

16 Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019:l4898.

17 Higgins JPT TJ, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.0 Cochrane; 2019 [Available from: www.training.cochrane.org/handbook.

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18 Furukawa TA, Barbui C, Cipriani A, et al. Imputing missing standard deviations in meta-analyses can provide accurate results. *J Clin Epidemiol* 2006;59:7–10.

19 Guyatt G, Oxman AD, Akl EA, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol* 2011;64:383–94.

20 Duke S-AS, Colagiuri S, Colagiuri R. Individual patient education for people with type 2 diabetes mellitus. *Cochrane Database Syst Rev* 2009;2009:CD005268.

21 McCall AL. Insulin therapy and hypoglycemia. *Endocrinol Metab Clin North Am* 2012;41:57–87.

22 Herman ME, O'Keefe JH, Bell DSH, et al. Insulin therapy increases cardiovascular risk in type 2 diabetes. *Prog Cardiovasc Dis* 2017;60:422–34.

23 Yardley JE, Hay J, Abou-Setta AM, et al. A systematic review and meta-analysis of exercise interventions in adults with type 1 diabetes. *Diabetes Res Clin Practice* 2014;106:393–400.

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Figure Legends

Figure 1. PRISMA Flow Diagram.

Figure 2. Risk of bias evaluation for each outcome. The risk of bias was assessed using the Cochrane risk of bias tool for randomized trials. TDD, total daily dose; LH, lipohypertrophy

Figure 3. Forest plots of primary and secondary outcomes

A. Change in total daily doses of insulin

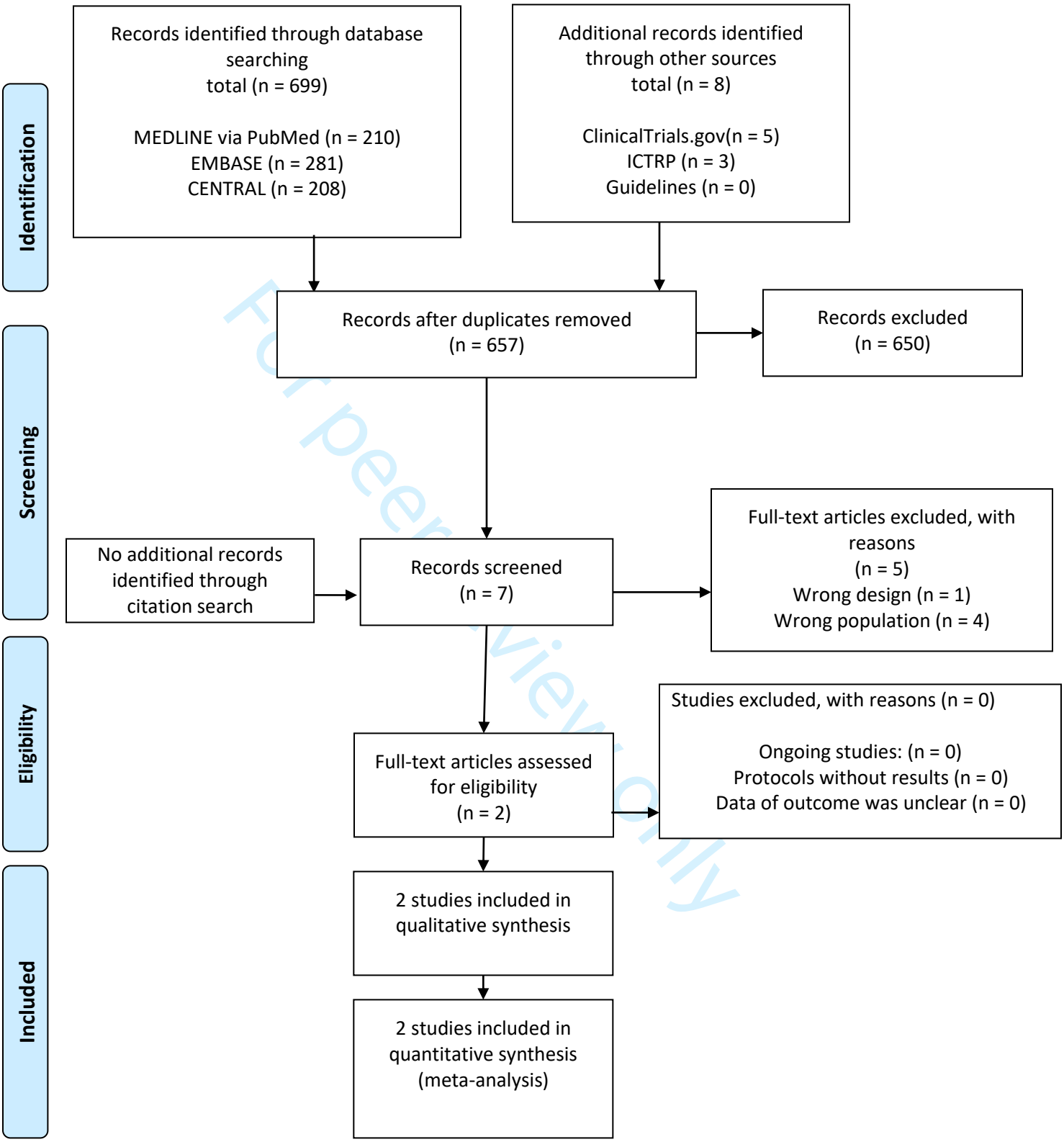
B. Change in HbA1c levels

C. Prevalence of hypoglycaemia

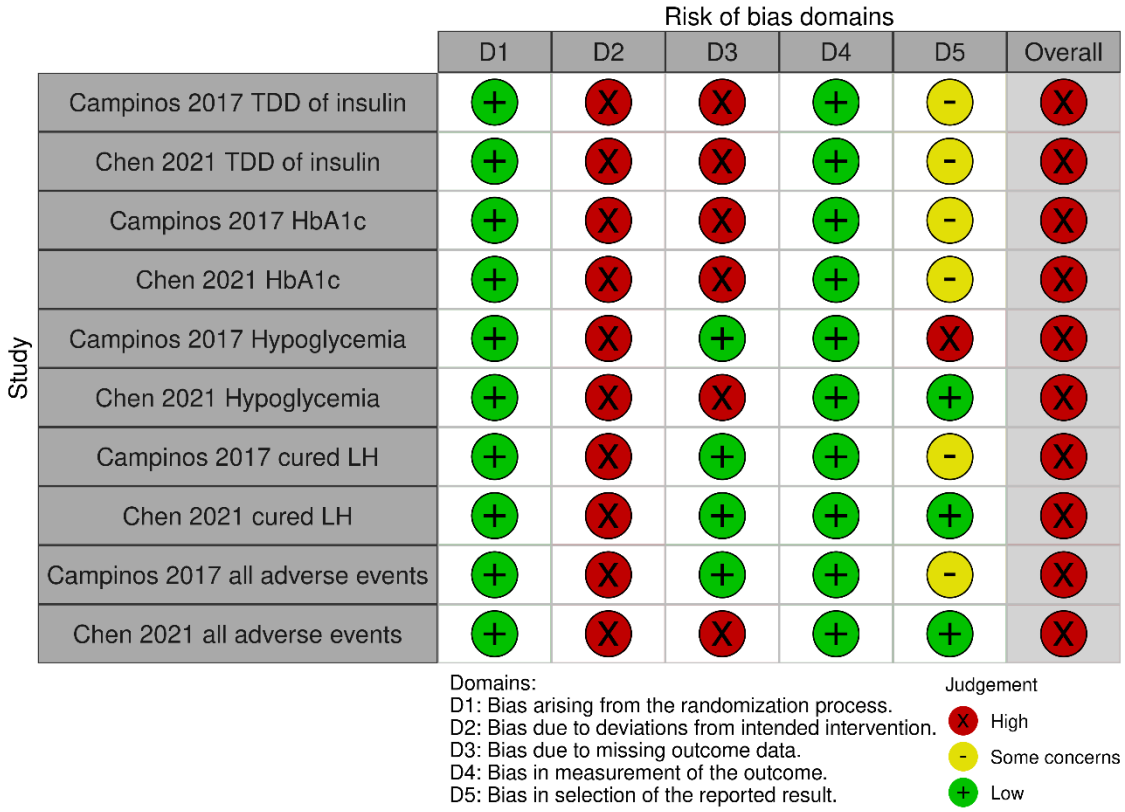
D. Change in the proportion of patients with cured LH

LH, lipohypertrophy

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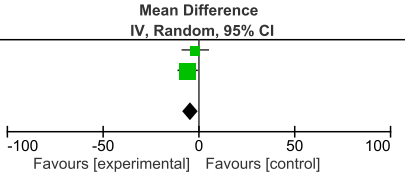

Risk of bias domains
D1
D2
D3
D4
D5
Overall
Study
Campinos 2017 TDD of insulin
Chen 2021 TDD of insulin
Campinos 2017 HbA1c
Chen 2021 HbA1c
Campinos 2017 Hypoglycemia
Chen 2021 Hypoglycemia
Campinos 2017 cured LH
Chen 2021 cured LH
Campinos 2017 all adverse events
Chen 2021 all adverse events
Domains:
D1: Bias arising from the randomization process.
D2: Bias due to deviations from intended intervention.
D3: Bias due to missing outcome data.
D4: Bias in measurement of the outcome.
D5: Bias in selection of the reported result.
Judgement
High
Some concerns
Low

eview only

A.

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Campinos 2017	-5.02	18.1952	53	-3.12	17.9167	56	34.2%	-1.90 [-8.68, 4.88]
Chen 2021	-6.8	19.1	106	-0.9	17	104	65.8%	-5.90 [-10.79, -1.01]
Total (95% CI)			**159**			**160**	**100.0%**	**-4.53 [-8.50, -0.57]**

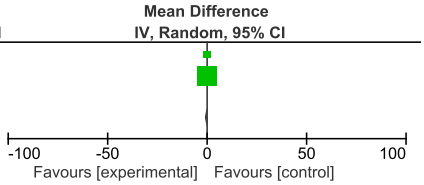
Heterogeneity: Tau² = 0.00; Chi² = 0.88, df = 1 (P = 0.35); I² = 0%
Test for overall effect: Z = 2.24 (P = 0.03)



B.

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Campinos 2017	-0.5	1.41	53	-0.54	1.42	56	20.1%	0.04 [-0.49, 0.57]
Chen 2021	-0.71	0.99	106	-0.55	0.98	104	79.9%	-0.16 [-0.43, 0.11]
Total (95% CI)			**159**			**160**	**100.0%**	**-0.12 [-0.36, 0.12]**

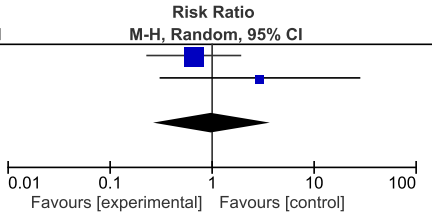
Heterogeneity: Tau² = 0.00; Chi² = 0.43, df = 1 (P = 0.51); I² = 0%
Test for overall effect: Z = 0.99 (P = 0.32)



C.

Study or Subgroup	Experimental Events	Total	Control Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
Campinos 2017	5	53	8	56	72.7%	0.66 [0.23, 1.89]
Chen 2021	3	106	1	104	27.3%	2.94 [0.31, 27.84]
Total (95% CI)		**159**		**160**	**100.0%**	**0.99 [0.27, 3.70]**
Total events	8		9			

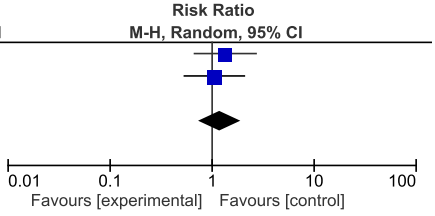
Heterogeneity: Tau² = 0.33; Chi² = 1.41, df = 1 (P = 0.23); I² = 29%
Test for overall effect: Z = 0.01 (P = 0.99)



D.

Study or Subgroup	Experimental Events	Total	Control Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
Campinos 2017	14	53	11	56	48.7%	1.34 [0.67, 2.69]
Chen 2021	15	106	14	104	51.3%	1.05 [0.53, 2.07]
Total (95% CI)		**159**		**160**	**100.0%**	**1.19 [0.73, 1.92]**
Total events	29		25			

Heterogeneity: Tau² = 0.00; Chi² = 0.25, df = 1 (P = 0.62); I² = 0%
Test for overall effect: Z = 0.69 (P = 0.49)



Supplementary Tables:
Table S1 PRISMA checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6, 7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	30, 31
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8, 9
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	10
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	10, 11
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	6, 7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary	11, 12

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Section and Topic	Item #	Checklist item	Location where item is reported
		statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	12
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	11, 12
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	13
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	13
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	13
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	12
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	14
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	14
Study characteristics	17	Cite each included study and present its characteristics.	14, 15, 16
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	17
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	17
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	14, 15
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	17
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	13
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	13
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	18
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	19
	23b	Discuss any limitations of the evidence included in the review.	20

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Section and Topic	Item #	Checklist item	Location where item is reported
	23c	Discuss any limitations of the review processes used.	21
	23d	Discuss implications of the results for practice, policy, and future research.	21
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	4, 6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	4, 6
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	13
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	22
Competing interests	26	Declare any competing interests of review authors.	22
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	22

Table S2 Search strategy

CENTRAL search strategy

#1 [mh Amyloidosis]
#2 lipohypertrophy:ti,ab
#3 "insulin-derived amyloidosis":ti,ab
#4 lipos:ti,ab
#5 "subcutaneous induration":ti,ab
#6 "subcutaneous nodules":ti,ab
#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8 [mh "Injections, Subcutaneous"]
#9 [mh "Patient Education as Topic"]
#10 injection*:ti,ab
#11 education*:ti,ab
#12 (reus*:ti,ab AND needle*:ti,ab)
#13 #8 OR #9 OR #10 OR #11 OR #12
#14 #7 AND #13

MEDLINE (via PubMed) search strategy

#1 "Amyloidosis"[Mesh]
#2 "lipohypertrophy"[tiab]
#3 "insulin-derived amyloidosis"[tiab]
#4 "lipos"[tiab]
#5 "subcutaneous induration"[tiab]
#6 "subcutaneous nodules"[tiab]
#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8 "Injections, Subcutaneous"[Mesh]

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#9 "Patient Education as Topic"[Mesh]
#10 injection*[tiab]
#11 education*[tiab]
#12 reus*[tiab] AND needle*[tiab]
#13 #8 OR #9 OR #10 OR #11 OR #12
#14 #7 AND #13
#15 randomized controlled trial[pt]
#16 controlled clinical trial[pt]
#17 randomized[tiab]
#18 placebo[tiab]
#19 drug therapy[sh]
#20 randomly[tiab]
#21 trial[tiab]
#22 groups[tiab]
#23 #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22
#24 animals[mh] NOT humans[mh]
#25 #23 NOT #24
#26 #14 AND #25

EMBASE (via ProQuest Dialog) search strategy
#1 emb(amyloidosis) OR ti(lipohypertrophy) OR ab(lipohypertrophy) OR ti(insulin-derived amyloidosis) OR ab(insulin-derived amyloidosis) OR ti(lipos) OR ab(lipos) OR ti(subcutaneous induration) OR ab(subcutaneous induration) OR ti(subcutaneous nodules) OR ab(subcutaneous nodules)
#2 exact(subcutaneous drug administration) OR exact(patient education) OR ti(injection*) OR ab(injection*) OR ti(education*) OR ab(education*) OR (ti(reus*) OR ab(reus*)) AND (ti(needle*) OR ab(needle*))
#3 (ab(random*) OR ti(random*)) OR (ab(clinical NEAR/1 trial*) OR ti(clinical NEAR/1 trial*)) OR (EMB.EXACT("health care quality"))
#4 #1 AND #2 AND #3

ICTRP search strategy
Participant: lipohypertrophy OR insulin-derived amyloidosis OR subcutaneous induration OR subcutaneous nodules
Intervention: injection OR education OR (reus AND needle)

ClinicalTrials.gov search strategy
Condition or disease: lipohypertrophy OR insulin-derived amyloidosis OR subcutaneous induration OR subcutaneous nodules
Intervention: injection OR education OR (reus AND needle)

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Table S3. The list of studies that were excluded from this review and the reason for exclusion

References	Reason for exclusion
Lori D et al. A Coloured Pen Needle Education System Improves Insulin Site Rotation Habits: Results of a Randomized Study. Diabetes Ther. 2020;11(12):2979-2991	Wrong population
Justin O et al. Nodular amyloidosis in a diabetic patient with frequent hypoglycemia: sequlae of repeatedly injecting insulin without site rotation. J Am Acad Dermatol. 2010;63(6):e113-114	Wrong study design
Inna V et al. A Randomized Controlled Trial to Assess the Impact of Proper Insulin Injection Technique Training on Glycemic Control. Diabetes Ther. 2017;8(6):1309-1318	Wrong population
Smith M et al. UK lipohypertrophy interventional study. Diabetes Res Clin Pract. 2017;126:248-253	Wrong population
Carina K et al. The roto track device improves rotation of insulin injections in T1DM: a proof-of-concept study. Diabetes. 2020;69(Supplement 1)	Wrong population

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BMJ Open

Efficacy of education on injection technique for patients diagnosed with diabetes with lipohypertrophy: a systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-055529.R1
Article Type:	Original research
Date Submitted by the Author:	07-Dec-2021
Complete List of Authors:	Ichikawa, Masahiro; Yokohama City University Medical Center, Department of Endocrinology and Diabetes Akiyama, Tomoaki; Yokohama City University Medical Center, Endocrinology and Diabetes Tsujimoto, Yasushi ; Kyoritsu Hospital, Nephrology and Dialysis Anan, Keisuke; Kyoto University, Department of Healthcare Epidemiology, Graduate School of Medicine and Public Health Yamakawa, Tadashi ; Yokohama City University Medical Center, Endocrinology and Diabetes Terauchi, Yasuo ; Yokohama City University School of Medicine Graduate School of Medicine, Endocrinology and Metabolism
Primary Subject Heading:	Diabetes and endocrinology
Secondary Subject Heading:	Diabetes and endocrinology
Keywords:	MEDICAL EDUCATION & TRAINING, DIABETES & ENDOCRINOLOGY, EDUCATION & TRAINING (see Medical Education & Training)

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Efficacy of education on injection technique for patients diagnosed with diabetes with lipohypertrophy: systematic review and meta-analysis

Masahiro Ichikawa[a]*, Tomoaki Akiyama[a], Yasushi Tsujimoto[b], Keisuke Anan[c], Tadashi Yamakawa[a], Yasuo Terauchi[d]

[a]Department of Endocrinology and Diabetes, Yokohama City University Medical Center, Yokohama, Japan

[b]Department of Nephrology and Dialysis, Kyoritsu Hospital, Kawanishi, Japan

[c]Department of Healthcare Epidemiology, Graduate School of Medicine and Public Health, Kyoto University, Kyoto, Japan

[d]Department of Endocrinology and Metabolism, Yokohama City University School of Medicine, Yokohama, Japan

***Corresponding Author**

Masahiro Ichikawa

Department of Endocrinology and Diabetes

Yokohama City University Medical Center

Urafunecho 4-57, Minami-ku

For peer review only

Yokohama 232-0024, Japan

Tel: +81-45-261-5656

Fax: +81-45-231-1846

E-mail: t196009c@yokohama-cu.ac.jp

Word count: 3879 words

Abstract: 299 words

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ABSTRACT

Objectives: This study aimed to investigate the efficacy of providing education on injection technique to patients with diabetes with lipohypertrophy (LH).

Design: We conducted a systematic review and meta-analysis.

Methods: We included patients with diabetes who use insulin and have LH, and excluded patients without LH. We performed a literature search on CENTRAL, MEDLINE, EMBASE, ICTRP, and ClinicalTrials.gov on November 2021 for randomised controlled trials (RCTs). We used the revised Cochrane Risk of Bias 2 tool to evaluate the risk of bias in each outcome in each study. We then pooled the data using a random-effects model and evaluated the certainty of evidence using the Grading of Recommendations Assessment, Development, and Evaluation approach.

Outcome measures: The primary endpoints were change in total daily dose (TDD) of insulin, change in HbA1c levels, and prevalence of hypoglycaemia.

Results: We screened 580 records and included three RCTs (637 participants) in the meta-analysis. Education on injection technique may slightly increase the change of TDD of insulin (3 studies, 637 participants: mean difference (MD) -6.26; 95% confidence interval (CI) -9.42, -3.10; $p < 0.001$; $I^2 = 38\%$; low certainty of evidence) and may have little to no effect on change in HbA1c but the evidence is very uncertain compared to that

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in the control group (three studies, 637 participants: MD -0.59; 95% CI -1.71, 0.54; p = 0.31; I^2 = 98%; very low certainty of evidence). Providing education about injection technique may have little-to-no effect on the prevalence of hypoglycaemia (three studies, 637 participants: risk ratio 0.44; 95% CI 0.06, 3.13; p = 0.41; I^2 = 90%; very low certainty of evidence).

Conclusions: The present meta-analysis suggests that injection technique education may result in a slight reduction in the TDD of insulin. However, the effect of education on HbA1c, hypoglycaemia, and cured LH is uncertain.

Protocol registration: DOI: dx.doi.org/10.17504/protocols.io.btiinkce

Keywords: injection technique, lipohypertrophy, diabetes mellitus

Strengths and limitations of this study

・To the best of our knowledge, this is the first systematic review and meta-analysis to investigate the effect of injection technique education in diabetes patients with lipohypertrophy.

・We conducted a comprehensive search and summarised the evidence for the effects of injection technique education, used a prospective registered protocol, followed the Cochrane Handbook and GRADE, selected a rigorous methodology, and adhered to reporting guidelines.

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- The limitation of this study is that we did not have sufficient data to perform a subgroup analysis.

INTRODUCTION

There has been a continuous increase in the number of diabetes patients worldwide, and diabetes has become a major public health problem, with 4.2 million adults estimated to die due to diabetes and its complications. [1] The number of diabetes patients is expected to increase to 700 million by 2045. [1] Blood glucose control is important to prevent the complications of diabetes, and insulin therapy plays a key role in regulating blood glucose level in patients. [2] Insulin therapy is necessary not only for type 1 diabetes but also for type 2 diabetes and is estimated to be used in 7.5% of type 2 diabetes patients. [3] Lipohypertrophy (LH), a side effect of long-term insulin injection, has been reported in 38% of diabetes patients. [4] LH is caused by the thickening of adipose tissue localised to the site of insulin injection. [5, 6]

Patients with LH face insulin variability because both insulin absorption and action becomes slow. [4, 7, 8] Moreover, the risk of unexpected hypoglycaemia has been noted in patients with LH. [7] Previous studies have suggested that providing proper education on injection techniques may reduce HbA1c levels and/or total daily dose (TDD) of insulin.

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[9-13] Injection techniques education (IT) includes rotating the injection sites properly, avoiding needle reuse, and avoiding injecting into the site of LH. [5] It is also recommended to use a 4-mm-long needle to avoid injection into the muscle. [5, 14]

To the best of our knowledge, there has been no systematic investigation on the effects of providing education about injection technique on the TDD of insulin, HbA1c levels, and hypoglycaemia in diabetes patients with LH. Therefore, we aimed to perform a systematic review and meta-analysis to determine the efficacy of providing education on the injection technique to diabetes patients with LH.

METHODS

Ethics approval

As this was a systematic review, there were no ethical issues involved. We registered the protocol in protocols.io (DOI: dx.doi.org/10.17504/protocols.io.btiinkce). Patient consent was not necessary for the present study.

Eligibility criteria

We conducted this systematic review in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, [15] and the checklist

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is shown in the online supplemental table S1. We conducted a systematic review investigating the effects of IT on patients with type 1 or type 2 diabetes who use insulin and have LH, and included individual, cluster, and crossover randomised trials. We did not limit the inclusion to languages or countries. We included published and unpublished papers, conference abstracts, letters, and all other types of papers. We excluded observational studies and non-randomised controlled trials (RCTs). We did not exclude studies because of the observation period or year of publication.

We included patients with type 1 or type 2 diabetes who had been receiving insulin injections for at least 1 year, with LH, 18 years old and above, any sex, no HbA1c restriction, and no education about LH within the past 6 months. LH diagnosis was made by a physician, nurse, or investigators when it was clinically visible, palpable, or visible on ultrasound. [4] We accepted any number of insulin injections per day or needle size for insulin injections.

We excluded pregnant women, patients who wished to become pregnant, lactating women, and patients taking medications that may cause LH (anti-retroviral or corticosteroid therapy).

Information sources

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Search strategy

We searched the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE using PubMed, and EMBASE using ProQuest Dialog on 11 November 2021. We also searched the World Health Organization International Clinical Trials Platform Search Portal (ICTRP) and ClinicalTrials.gov on 11 November 2021 for ongoing or unpublished trials. The search strategies are presented in the online supplemental Table S2. We checked the reference lists of the studies, including international guidelines, and identified the reference lists of eligible studies and articles citing eligible studies.

Selection process

Two independent reviewers (MI and TA) screened the titles and abstracts, followed by an assessment of eligibility based on the full texts. We contacted the original authors if the relevant data were missing. Disagreements between the two reviewers were resolved by discussion, and if this was not possible, a third reviewer (YT or KA) became an arbiter.

Data collection process

Two reviewers (MI and TA) independently extracted data from included studies using a standardised data collection form. Disagreements were resolved by discussion, and if this

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was not possible, a third reviewer (YT or KA) became an arbiter. We conducted up to two author queries by e-mail to obtain relevant data from the researchers.

Data items

The primary endpoints were changes in insulin TDD and HbA1c levels, and the prevalence of hypoglycaemia. The change in insulin TDD was defined as the change in TDD from baseline, and the duration was the longest follow-up period after 3 months. The change in HbA1c levels was defined as the change in HbA1c from baseline, and the duration was the longest follow-up period after 3 months. Hypoglycaemia was defined as one or more hypoglycaemic symptoms (palpitations, fatigue, sweating, strong hunger, dizziness, tremors, etc.) and a blood glucose level below 60 mg/dL. [7] We accepted hypoglycaemia as defined by the original authors. Duration was the longest follow-up period after 3 months. The secondary endpoints were changes in the proportion of patients with cured LH and all adverse events. The definition of change in the proportion of patients with cured LH was the same as that of the original paper, and the duration was the longest follow-up period after 3 months. All adverse events were the same as those defined in the original study.

The data collecting form included information on the first author's name, year of

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publication, country, sample size, proportion of male participants, mean age of participants, number of participants with type 1 diabetes mellitus (T1DM)/type 2 diabetes mellitus (T2DM), estimated diabetes duration, mean duration of insulin treatment, intervention details, detection methods of LH, and the outcome measures mentioned above. We asked the authors of the original papers for unpublished or additional data and obtained research data for [13] but not for [12].

Study risk of bias assessment

Two reviewers (MI and TA) independently assessed the risk of bias using the Risk of Bias 2 (RoB2) tool. [16] The effect of interest is the intention-to-treat effect, which is the effect assigned to the interventions at baseline, regardless of whether or not the intervention was received as intended. Disagreements between the two reviewers were discussed, and if this was not possible, a third reviewer (YT or KA) became an arbiter.

Effect measures

We pooled the relative risk ratios and 95% confidence intervals (CIs) for the following binary variables: hypoglycaemia and proportion of patients with cured LH. We pooled the mean differences and 95% CIs for the following continuous variables: HbA1c and

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TDD of insulin.

We extracted the data on an intention-to-treat basis for all dichotomous data whenever possible. For continuous data, we did not impute missing data based on the recommendation of the Cochrane Handbook. [17] We performed a meta-analysis of the data available in the original study.

When the original studies reported only standard errors or p-values, we used Altman's method to calculate the standard deviation. [18] When these values were not available after enquiring the authors, we calculated the standard deviations using confidence intervals and t-values based on the methods in the Cochrane Handbook [17] or validated methods. [18] We analysed the validity of these methods using sensitivity analysis.

Statistical analyses

We used Review Manager software (RevMan 5.4; The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark) to conduct a meta-analysis using the DerSimonian and Laird random-effects method. We assessed statistical heterogeneity by visual inspection of the forest plots and calculation of I^2 statistic (I^2 values of 0-40%: might not be important; 30-60%: may represent moderate heterogeneity; 50-90%: may represent substantial heterogeneity; 75-100%, considerable heterogeneity). For the I^2

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statistic, we performed the Cochrane Chi^2 test (Q test) and considered it statistically significant if the P value was less than 0.10.

Certainty assessment

Based on the Cochrane Handbook, [17] we created a summary of findings (SoF) table for the results of changes in the insulin TDD, changes in HbA1c levels, and the prevalence of hypoglycaemia. A SoF table contained important information on the relative and absolute effect sizes of the interventions investigated, the amount of available evidence, and the certainty of available evidence. [17]

We assessed the quality of evidence for each outcome based on the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) method. [19] GRADE specifies four categories, namely high, moderate, low, and very low, which are applied to accumulating evidence. Evidence from randomised trials starts with high certainty of evidence and can be downgraded by five domains: risk of bias, imprecision, inconsistency, indirectness, and publication bias.

Difference between a protocol and a review

The duration of hypoglycemia was defined as the follow-up period, but as it was reported

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in terms of the number of patients who had hypoglycemia, the duration was defined as the longest follow-up after 3 months. We could not summarize all adverse events because there were few reports other than hypoglycaemia. To clarify the influence of effect modifiers on the results, subgroup analyses of the primary outcome were planned to be evaluated with the following factors when sufficient data were available: age (< 65 years vs. ≥ 65 years), baseline HbA1c level of < 8% vs. ≥ 8%, [20] sex, and IT protocol with or without initial insulin level reduction.

We also planned to conduct the following sensitivity analyses on the primary outcomes to assess whether the results of the review were robust to the decisions made during the review process: exclusion of studies with imputed statistics, and exclusion of studies with an overall high risk of bias. We could not perform a predetermined subgroup analysis and sensitivity analysis because of insufficient data.

We planned to assess the possibility of publication bias by visually inspecting the funnel plots and testing the Egger's test. If there were fewer than 10 trials or trials with similar sample sizes, then we decided not to perform the test. We could not perform the Egger test for reporting bias detection because there were fewer than 10 trials.

Patients and public involvement

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As this was a systematic review, patients or the public were not involved in our research.

RESULTS

Study selection

We performed the search in November 2021 and found 580 records; after duplicates were removed, 550 records were screened for titles and abstracts. Nine full-text records were assessed for eligibility. Six studies were excluded, and the reasons were as follows: five for including the wrong population and one for using the wrong design. Details of the excluded studies are shown in the online supplemental Table S3. Finally, we included three RCTs that met the eligibility criteria (Figure 1). [12, 13, 21]

Study characteristics

Table 1 presents the characteristics of the three included studies conducted in France, China, and Italy. In the French study, the number of individuals with T1DM/T2DM was 66/57, and in the Italian study, it was 0/318. In the Chinese study, there were fewer than seven patients with T1DM, but the exact number of patients with T1DM/T2DM could not be obtained despite inquiries. In the French study, all participants received information about optimal IT and LH at baseline for ethical reasons. In the Italian study,

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the proportion of cured LH was not reported.

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Table 1. Characteristics of the included studies.

Study	Methods	Subject characteristics	Intervention details	Reported outcome of interest
Campions 2017 [12]	Location: France Sample size: 123 Detection method of LH: visible and/or palpable LH determined by nurses	Age (years): 52.8 ± 15.1 (IT), 51.4 ± 16.4 (C) Male (%): 70.7 Number of T1DM/T2DM individuals: 66/57 Estimated diabetes duration (years): 19.8 ± 11.5 (IT), 16.3 ± 9.2 (C) Insulin treatment duration (years): 14.8 ± 12.8 (IT), 12.5 ± 10 (C)	IT: reducing insulin doses initially by 20%, instructing to rotate within injection sites, foregoing needle reuse, stopping injecting into the LH site and switching to 4-mm needles. C: standard care. All participants received information about optimal IT and LH at baseline.	Change in TDD of insulin from baseline Change in HbA1c levels from baseline Hypoglycaemia Change in the proportion of patients with cured LH All adverse events
Chen 2021 [13]	Location: China Sample size: 210 Detection method of LH: visible and/or palpable LH determined by nurses and ultrasound examination	Age (years): 59 ± 7 (IT), 60 ± 9 (C) Male (%): 46.7 Number of T1DM/T2DM individuals: N/A Estimated diabetes duration (years): 14.4 ± 5.8 (IT), 16.8 ± 7.6 (C) Insulin treatment duration (years): 6.9 ± 4.7 (IT), 7.1 ± 5.4 (C)	IT: reducing insulin doses initially by 20%, instructing to rotate within injection sites, foregoing needle reuse, stopping injecting into the LH site and switching to 4-mm needles. C: standard care	Change in TDD of insulin from baseline Change in HbA1c levels from baseline Hypoglycaemia Change in the proportion of patients with cured LH All adverse events

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Gentile 2021 [21]	Location: Italy Sample size: 318 Detection method of LH: inspection and palpation by healthcare professionals and ultrasound-based skin evaluations	Age (years): 61 ± 10 (IT), 63 ± 12 (C) Male (%): 44.0 Number of T1DM/T2DM individuals: 0/318 Estimated diabetes duration (years): 11.6 ± 9.8 (IT), 11.3 ± 5.7 (C) Insulin treatment duration (years): 6.5 ± 9.3 (IT), 6.7 ± 7.2 (C)	IT: reducing insulin doses initially by 20%, repeatedly instructing to rotate within injection sites, foregoing needle reuse and stopping injecting into the LH site. Change to 4 mm needles 6 months before randomization. Insulin titration. C: reducing insulin doses initially by 20%, at the first session instructing to rotate within injection sites, foregoing needle reuse and stopping injecting into the LH site. Change to 4 mm needles 6 months before randomization. Insulin titration.	Change in TDD of insulin from baseline Change in HbA1c levels from baseline Hypoglycaemia All adverse events

DM, Diabetes mellitus; IT, Injection technique education group; C, Control group; LH, lipohypertrophy; TDD, Total daily doses; N/A, Not available

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Risk of bias in the studies

Figure 2 shows the RoB2 result: In two of three RCTs, [12, 13] domain 2 (bias due to deviations from the intended intervention) and/or domain 3 (bias due to missing outcome data) were at high risk, and the overall risk of bias was high.

Outcome and certainty of evidence

Table 2 shows a summary of these findings. Figure 3 shows a forest plot. Compared with standard care, IT slightly increased the change in TDD of insulin (three studies, 637 participants: MD -6.26; 95% confidence interval (CI) -9.42, -3.10; p <0.001; I^2 = 38%; low certainty of evidence). The certainty of evidence was low because of serious risk of bias and imprecision. Compared with standard care, IT may increase little to no effect on the change in HbA1c levels but the evidence is very uncertain (three studies, 637 participants: MD -0.59; 95% CI -1.71, 0.54; p = 0.31; I^2 = 98%; very low certainty of evidence). The certainty of evidence was very low because of serious risk of bias, imprecision, and inconsistency.

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Table 2. Summary of findings comparing indicated outcomes between injection technique education and standard care.

Patient or population: Patients with type 1 diabetes or type 2 diabetes who use insulin and have LH

Setting: Outpatient

Intervention: Injection techniques education

Comparison: Standard care

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Quality of evidence (GRADE)
	Assumed risk-C	Corresponding risk-IT			
Change in TDD of insulin	Mean change in the TDD of insulin in the C group was -3.12 to -0.9.	Mean change in the TDD of insulin in the IT group was 6.26 lower(9.42 lower to 3.10 lower)		637 (3)	Low[a,b]
Change in HbA1c level	Mean change in HbA1c level in the C group was -0.55 to 0.1.	Mean change in HbA1c level in the IT group was 0.59 lower(1.71 lower to 0.54 higher)		637 (3)	Very low[a,b,c]
Prevalence of hypoglycaemia	162/319 (50.8%)	21/318 (6.60%)	RR 0.44 (0.06 to 3.13)	637 (3)	Very low[a,c,d]

* Corresponding risk (and 95% CI) are based on assumed risks in the control group and relative risks (and 95% CI) of the intervention. [a] Due to serious risk of bias
[b] Due to serious imprecision. The sample size did not meet the optimal size criterion.
[c] Due to serious inconsistency
[d] Due to very serious imprecision. The sample size did not meet the optimal size criterion, and the 95% CI was wide.LH, lipohypertrophy; C, Control; IT, Injection techniques education; TDD, total daily doses; HbA1c, haemoglobin A1c; RR, risk ratio; CI, confidence interval. GRADE Working Group grades of evidence: High quality: we are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect. Very low quality: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

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The evidence is very uncertain about the effect of insulin injection technique on the prevalence of hypoglycaemia (three studies, 637 participants: risk ratio 0.44; 95% CI 0.06, 3.13; p = 0.41; I^2 = 90%; very low certainty of evidence). The very low certainty of evidence was due to the serious risk of bias and inconsistency and the very serious imprecision. The evidence is very uncertain about the prevalence of cured LH compared to the control group (two studies, 319 participants: RR 1.19; 95% CI 0.73, 1.92; p = 0.49; I^2 = 0%; very low certainty of evidence). The very low certainty of evidence was due to the serious risk of bias and very serious imprecision.

DISCUSSION

The present study showed that IT may result in a slight reduction in the TDD of insulin. The evidence is very uncertain about the effect of insulin injection technique on the change in HbA1c levels, the prevalence of hypoglycaemia and the prevalence of cured LH compared to those in the control group. To the best of our knowledge, this is the first systematic review and meta-analysis to study the effects of insulin IT in comparison with standard care for diabetes patients with LH.

The results of the present study were partly consistent with those of previous observational studies, which found that IT was associated with changes in TDD of insulin and HbA1c levels. [9, 10] These previous observational studies reported that IT decreased the TDD of insulin and HbA1c levels in patients with diabetes upon insulin injection. Our meta-analysis of randomised trials showed similar results of TDD of insulin. Meanwhile, the results of the present study did not show a decrease in HbA1c levels, but this might be due to the fact that the three studies included were initially conducted with up to 20% reduction in TDD of insulin to prevent hypoglycaemia.

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Excessive exogenous insulin can be harmful, and it is important to reduce the TDD of insulin to prevent adverse events. Excessive exogenous insulin administration has been reported to be associated with adverse events such as hypoglycaemia, weight gain, and increased cardiovascular events and mortality. [22, 23] A meta-analysis in type 1 diabetes patients integrated aerobic-only intervention studies in adults, and multiple types of exercise interventions, including aerobic and anaerobic, in 12- to 18-year-olds. The results showed a reduction of 0.21 U/kg in TDD of insulin. [24] In the present study, the TDD of insulin was reduced by 6.26 U/day, which is less than that obtained with exercise intervention, which is effective in avoiding excessive insulin administration.

The certainty of evidence in the present study was low or very low. In one included study, [13] a portion of the control group was shifted to the ideal injection method, which contributed to the effect of high risk of bias due to deviation from the intended intervention. In the other two studies, some of the control groups was shifted to the ideal injection method. [12, 21] Despite the fact that some of the control group shifted to the ideal injection method, the present study showed a slight reduction in the TDD of insulin, which underestimates the effect. In fact, the effect may be even greater. In two of three studies, [12, 13] the bias due to missing outcome data for the continuous variable outcome was high because the missing value was greater than 5%. The bias due to missing outcome data for the binary variable outcome was low or high for the frequency of outcome and number of missing outcomes. The registration of the prevalence of hypoglycaemia in clinicaltrials.gov mentions only a timeframe of 6 months, but because the prevalence of hypoglycaemia was counted at 3 and 6 months, we considered the bias in the selection of the [12] as high. The other reason for lowering the GRADE is imprecision, because only three studies were included, and the sample size was small. In the future, more studies

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with sufficient sample sizes are needed.

The strengths of this study are that we conducted a comprehensive search and summarised the evidence for the effects of IT, used a prospective registered protocol, followed the Cochrane Handbook and GRADE, selected a rigorous methodology, and adhered to reporting guidelines.

However, this study had several limitations. First, as described above, we could not have enough data to conduct subgroup analyses and assess publication bias.It was not plausible for publication bias to affect our findings because our comprehensive search did not show any ongoing or unpublished studies. Second, the generalisability of our findings are limited. For example, the included studies tended to recruit middle-aged, obese, and patients with type 2 diabetes patients, and measured the outcomes at 6 months. However, a crossover study limited to type 1 diabetes individuals has shown that injections administered at sites away from the LH improved insulin absorption and reduce glycaemic variability compared to injections administered at the LH. [8] Previous studies have also shown the efficacy of optimised injection technique on TDD of insulin, although baseline characteristics varied, suggesting that it may be effective in the aforementioned populations. [9, 10] Further studies evaluating different populations and long-term outcomes are required. Third, we could not perform a subgroup analysis of types of DM which was requested during peer-review due to the limited data. The prevalence of LH was higher in type 2 diabetes than in type 1 diabetes, thus the types of DM may be an effect modifier of IT. [4] Future studies that use individual participant data can assess the effect modification.

CONCLUSION

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The present meta-analysis suggests that providing education on injection technique to diabetes patients with LH may slightly reduce the TDD of insulin. Clinicians may consider providing IT to patients with diabetes with LH, but they also need to understand the uncertainty of the evidence. In the future, we hope that RCTs with less bias, especially for the deviation of the intended protocol and with a rigorous protocol, will be conducted to accurately evaluate the effects.

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COMPETING INTERESTS:

All authors have no conflicts of interest to declare.

AUTHORS' CONTRIBUTIONS

MI conducted the search and analysed the data. MI and TA conducted screening of titles and abstracts, extracted data, and evaluated the risk of bias. If the disagreement could not be resolved, then YT or KA became an arbiter. MI, YT, and KA drafted the manuscript. YT, KA, TY, and YT revised the manuscript. All authors approved the manuscript version submitted for publication and agreed to be responsible for all aspects of this work.

ETHICAL APPROVAL: Not required.

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PATIENT CONSENT FORM

Not required

DATA SHARING STATEMENT The dataset is available from the corresponding author upon request.

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References

1 *Federation ID. IDF diabetes atlas*. 9th ed. http://www.diabetesatlas.org/ (Accessed 11 Jun 2021).

2 Brown A, Guess N, Dornhorst A, et al. Insulin-associated weight gain in obese type 2 diabetes mellitus patients: What can be done? *Diabetes Obes Metab* 2017;**19**:1655–68.

3 Basu S, Yudkin JS, Kehlenbrink S, et al. Estimation of global insulin use for type 2 diabetes, 2018–30: A microsimulation analysis. *Lancet Diabetes Endocrinol* 2019;**7**:25–33.

4 Deng N, Zhang X, Zhao F, et al. Prevalence of lipohypertrophy in insulin-treated diabetes patients: A systematic review and meta-analysis. *J Diabetes Investig* 2017;**9**:536–43.

5 Frid AH, Kreugel G, Grassi G, et al. New insulin delivery recommendations. *Mayo Clin Proc* 2016;**91**:1231–55.

6 Gentile S, Guarino G, Corte TD, et al. Insulin-induced skin lipohypertrophy in type 2 diabetes: A multicenter regional survey in Southern Italy. *Diabetes Ther*

2020;**11**:2001–17.

7 Blanco M, Hernández MT, Strauss KW, et al. Prevalence and risk factors of lipohypertrophy in insulin-injecting patients with diabetes. *Diabetes Metab* 2013;**39**:445–53.

8 Famulla S, Hövelmann U, Fischer A, et al. Insulin injection into lipohypertrophic tissue: Blunted and more variable insulin absorption and action and impaired postprandial glucose control. *Diabetes Care* 2016;**39**:1486–92.

9 Grassi G, Scuntero P, Trepiccioni R, et al. Optimizing insulin injection technique and its effect on blood glucose control. *J Clin Transl Endocrinol* 2014;**1**:145–50.

10 Smith M, Clapham L, Strauss K. UK lipohypertrophy interventional study. *Diabetes Res Clin Pract* 2017;**126**:248–53.

11 Misnikova IV, Gubkina VA, Lakeeva TS, et al. A randomized controlled trial to assess the impact of proper insulin injection technique training on glycemic control. *Diabetes Ther* 2017;**8**:1309–18.

12 Campinos C, Le Floch JP, Petit C, et al. An effective intervention for diabetic lipohypertrophy: Results of a randomized, controlled, prospective multicenter study in France. *Diabetes Technol Ther* 2017;**19**:623–32.

13 Chen L, Xing Q, Li J, et al. Injection technique education in patients with diabetes injecting insulin into areas of lipohypertrophy: A randomized controlled trial. *Diabetes Ther* 2021. doi: 10.1007/s13300-021-01013-1 [Published online first: 2021/02/12].

14 Frid AH, Hirsch LJ, Menchior AR, et al. Worldwide injection technique questionnaire study: Injecting complications and the role of the professional. *Mayo Clin Proc* 2016;**91**:1224–30.

For peer review only

15 Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71.

16 Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;366:l4898.

17 Higgins JPT TJ. In: Chandler J, Cumpston M, Li T, Page MJ, Welch VA, (editors). http://www.training.cochrane.org/handbook. *Cochrane Handbook for Systematic Reviews of Interventions version 6.0 Cochrane* 2019.

18 Furukawa TA, Barbui C, Cipriani A, et al. Imputing missing standard deviations in meta-analyses can provide accurate results. *J Clin Epidemiol* 2006;**59**:7–10.

19 Guyatt G, Oxman AD, Akl EA, et al [GRADE guidelines]. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol* 2011;**64**:383–94.

20 Duke SA, Colagiuri S, Colagiuri R. Individual patient education for people with type 2 diabetes mellitus. *Cochrane Database Syst Rev* 2009;(1):CD005268.

21 Gentile S, Guarino G, Della Corte T, et al. Role of structured education in reducing Lypodistrophy and its metabolic complications in insulin-treated people with Type 2 diabetes: A randomized multicenter case-control study. *Diabetes Ther* 2021;**12**:1379–98.

22 McCall AL. Insulin therapy and hypoglycemia. *Endocrinol Metab Clin North Am* 2012;**41**:57–87.

23 Herman ME, O'Keefe JH, Bell DSH, et al. Insulin therapy increases cardiovascular risk in type 2 diabetes. *Prog Cardiovasc Dis* 2017;**60**:422–34.

24 Yardley JE, Hay J, Abou-Setta AM, et al. A systematic review and meta-analysis of exercise interventions in adults with type 1 diabetes. *Diabetes Res Clin Pract*

For peer review only

2014;**106**:393–400.

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Figure Legends

Figure 1. PRISMA Flow Diagram.

Figure 2. Risk of bias evaluation for each outcome. The risk of bias was assessed using the Cochrane risk of bias tool for randomised trials. TDD, total daily dose; LH, lipohypertrophy

Figure 3. Forest plots of primary and secondary outcomes

A. Change in total daily doses of insulin

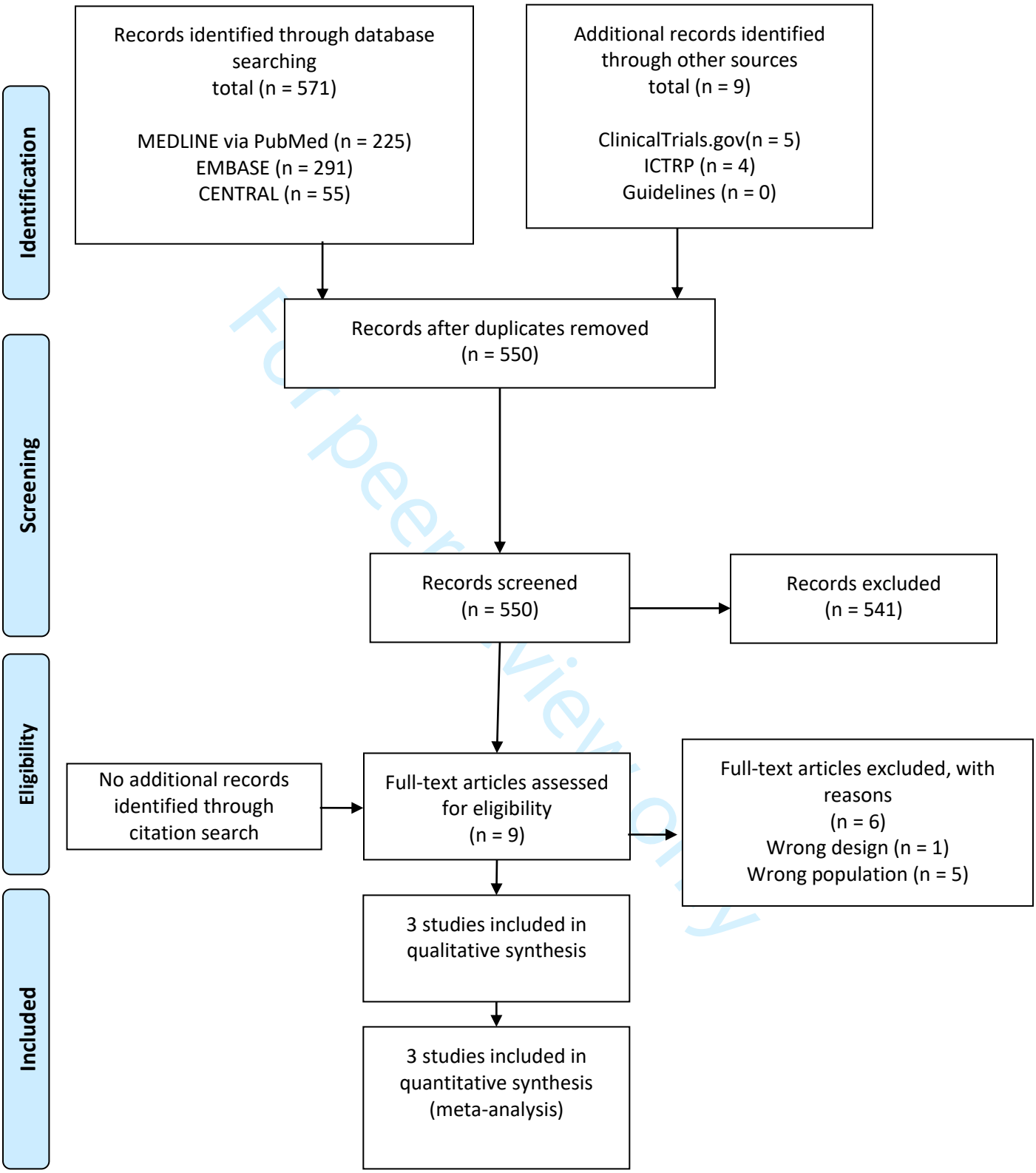
B. Change in HbA1c levels

C. Prevalence of hypoglycaemia

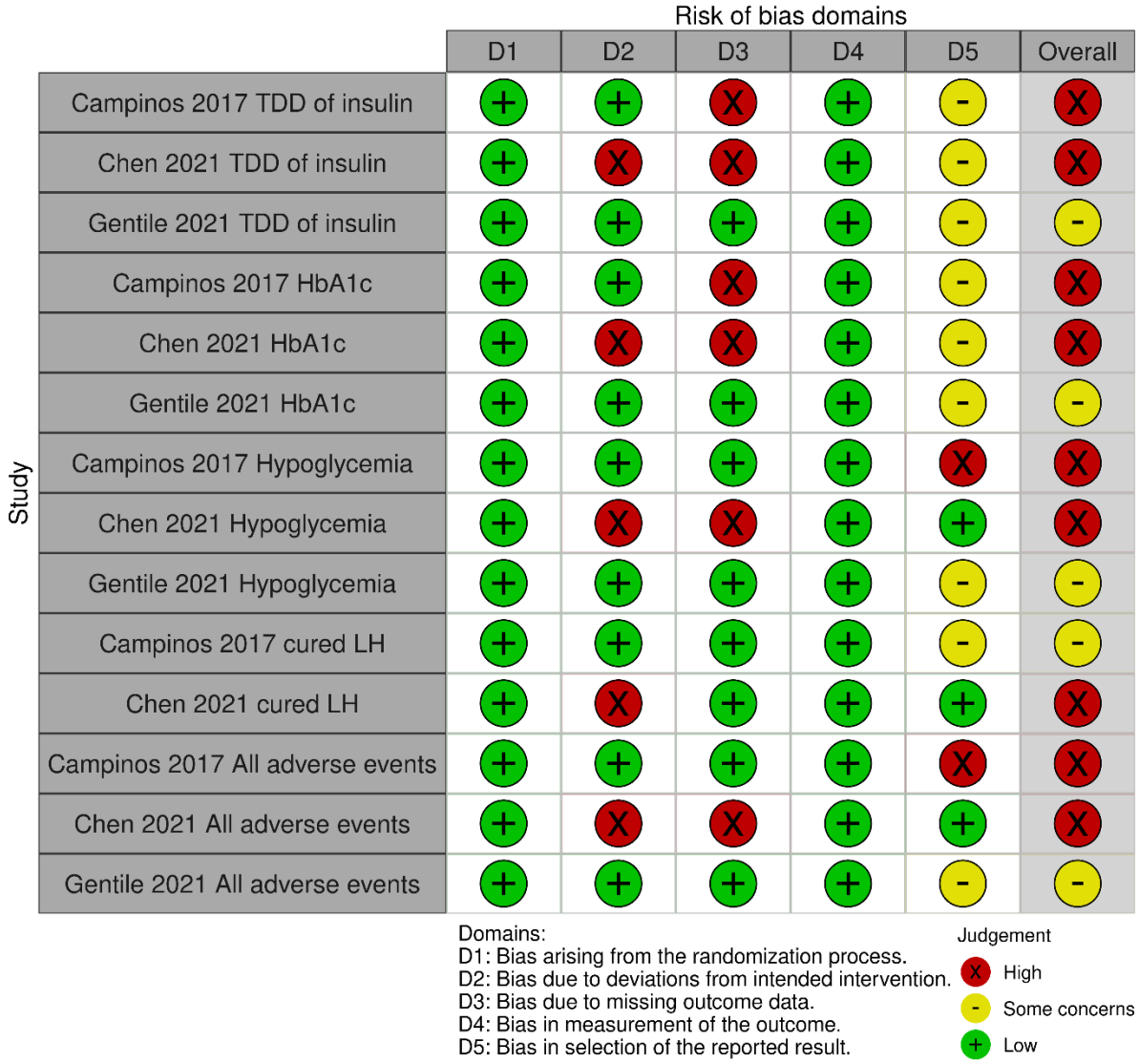
D. Change in the proportion of patients with cured LH

LH, lipohypertrophy

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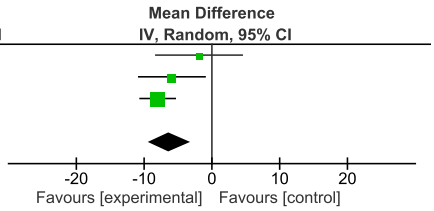
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Risk of bias domains
D1
D2
D3
D4
D5
Overall
Study
Campinos 2017 TDD of insulin
Chen 2021 TDD of insulin
Gentile 2021 TDD of insulin
Campinos 2017 HbA1c
Chen 2021 HbA1c
Gentile 2021 HbA1c
Campinos 2017 Hypoglycemia
Chen 2021 Hypoglycemia
Gentile 2021 Hypoglycemia
Campinos 2017 cured LH
Chen 2021 cured LH
Campinos 2017 All adverse events
Chen 2021 All adverse events
Gentile 2021 All adverse events
Domains:
D1: Bias arising from the randomization process.
D2: Bias due to deviations from intended intervention.
D3: Bias due to missing outcome data.
D4: Bias in measurement of the outcome.
D5: Bias in selection of the reported result.
Judgement
High
Some concerns
Low

A.

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Campinos 2017	-5.02	16.9065	53	-3.12	16.9902	56	19.0%	-1.90 [-8.27, 4.47]
Chen 2021	-6.8	19.1	106	-0.9	17	104	27.8%	-5.90 [-10.79, -1.01]
Gentile 2021	46	9	159	54	14	159	53.2%	-8.00 [-10.59, -5.41]
Total (95% CI)			**318**			**319**	**100.0%**	**-6.26 [-9.42, -3.10]**

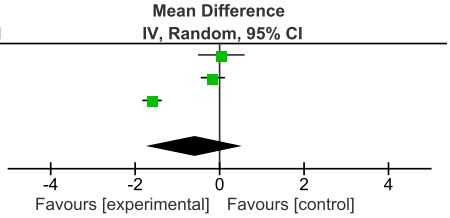
Heterogeneity: Tau² = 3.14; Chi² = 3.23, df = 2 (P = 0.20); I² = 38%
Test for overall effect: Z = 3.88 (P = 0.0001)



B.

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Campinos 2017	-0.5	1.41	53	-0.54	1.42	56	32.0%	0.04 [-0.49, 0.57]
Chen 2021	-0.71	0.99	106	-0.55	0.98	104	33.9%	-0.16 [-0.43, 0.11]
Gentile 2021	6.6	0.9	159	8.2	1	159	34.1%	-1.60 [-1.81, -1.39]
Total (95% CI)			**318**			**319**	**100.0%**	**-0.59 [-1.71, 0.54]**

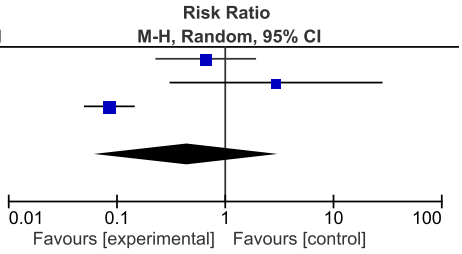
Heterogeneity: Tau² = 0.96; Chi² = 84.21, df = 2 (P < 0.00001); I² = 98%
Test for overall effect: Z = 1.02 (P = 0.31)



C.

Study or Subgroup	Experimental Events	Total	Control Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
Campinos 2017	5	53	8	56	35.5%	0.66 [0.23, 1.89]
Chen 2021	3	106	1	104	26.0%	2.94 [0.31, 27.84]
Gentile 2021	13	159	153	159	38.5%	0.08 [0.05, 0.14]
Total (95% CI)		**318**		**319**	**100.0%**	**0.44 [0.06, 3.13]**
Total events	21		162			

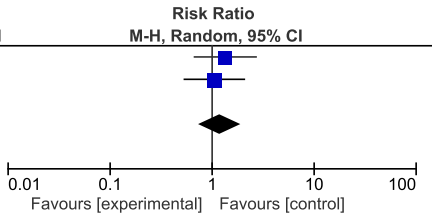
Heterogeneity: Tau² = 2.52; Chi² = 19.11, df = 2 (P < 0.0001); I² = 90%
Test for overall effect: Z = 0.82 (P = 0.41)



D.

Study or Subgroup	Experimental Events	Total	Control Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
Campinos 2017	14	53	11	56	48.7%	1.34 [0.67, 2.69]
Chen 2021	15	106	14	104	51.3%	1.05 [0.53, 2.07]
Total (95% CI)		**159**		**160**	**100.0%**	**1.19 [0.73, 1.92]**
Total events	29		25			

Heterogeneity: Tau² = 0.00; Chi² = 0.25, df = 1 (P = 0.62); I² = 0%
Test for overall effect: Z = 0.69 (P = 0.49)



Supplementary Tables:
Table S1 PRISMA checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6, 7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table S2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8, 9
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	9, 10
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	10, 11
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	6, 7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary	11, 12

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Section and Topic	Item #	Checklist item	Location where item is reported
		statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	12
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	11, 12
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	13
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	13
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	13
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	12
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	14
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	14
Study characteristics	17	Cite each included study and present its characteristics.	14, 15, 16
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	18, Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	18, 20
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	14, 15, 16, 18
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	18, 20, Figure 3
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	13
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	13
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	20
	23b	Discuss any limitations of the evidence included in the review.	22

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Section and Topic	Item #	Checklist item	Location where item is reported
	23c	Discuss any limitations of the review processes used.	22
	23d	Discuss implications of the results for practice, policy, and future research.	23
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	4, 6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	4, 6
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	13
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	23
Competing interests	26	Declare any competing interests of review authors.	23
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	24

Table S2 Search strategy

CENTRAL search strategy

#1 [mh Amyloidosis]
#2 lipohypertrophy:ti,ab
#3 "insulin-derived amyloidosis":ti,ab
#4 lipos:ti,ab
#5 "subcutaneous induration":ti,ab
#6 "subcutaneous nodules":ti,ab
#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8 [mh "Injections, Subcutaneous"]
#9 [mh "Patient Education as Topic"]
#10 injection*:ti,ab
#11 education*:ti,ab
#12 (reus*:ti,ab AND needle*:ti,ab)
#13 #8 OR #9 OR #10 OR #11 OR #12
#14 #7 AND #13

MEDLINE (via PubMed) search strategy

#1 "Amyloidosis"[Mesh]
#2 "lipohypertrophy"[tiab]
#3 "insulin-derived amyloidosis"[tiab]
#4 "lipos"[tiab]
#5 "subcutaneous induration"[tiab]
#6 "subcutaneous nodules"[tiab]
#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8 "Injections, Subcutaneous"[Mesh]

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#9 "Patient Education as Topic"[Mesh]
#10 injection*[tiab]
#11 education*[tiab]
#12 reus*[tiab] AND needle*[tiab]
#13 #8 OR #9 OR #10 OR #11 OR #12
#14 #7 AND #13
#15 randomized controlled trial[pt]
#16 controlled clinical trial[pt]
#17 randomized[tiab]
#18 placebo[tiab]
#19 drug therapy[sh]
#20 randomly[tiab]
#21 trial[tiab]
#22 groups[tiab]
#23 #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22
#24 animals[mh] NOT humans[mh]
#25 #23 NOT #24
#26 #14 AND #25

EMBASE (via ProQuest Dialog) search strategy
#1 emb(amyloidosis) OR ti(lipohypertrophy) OR ab(lipohypertrophy) OR ti(insulin-derived amyloidosis) OR ab(insulin-derived amyloidosis) OR ti(lipos) OR ab(lipos) OR ti(subcutaneous induration) OR ab(subcutaneous induration) OR ti(subcutaneous nodules) OR ab(subcutaneous nodules)
#2 exact(subcutaneous drug administration) OR exact(patient education) OR ti(injection*) OR ab(injection*) OR ti(education*) OR ab(education*) OR (ti(reus*) OR ab(reus*)) AND (ti(needle*) OR ab(needle*))
#3 (ab(random*) OR ti(random*)) OR (ab(clinical NEAR/1 trial*) OR ti(clinical NEAR/1 trial*)) OR (EMB.EXACT("health care quality"))
#4 #1 AND #2 AND #3

ICTRP search strategy
Participant: lipohypertrophy OR insulin-derived amyloidosis OR subcutaneous induration OR subcutaneous nodules
Intervention: injection OR education OR (reus AND needle)

ClinicalTrials.gov search strategy
Condition or disease: lipohypertrophy OR insulin-derived amyloidosis OR subcutaneous induration OR subcutaneous nodules
Intervention: injection OR education OR (reus AND needle)

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Table S3. The list of studies that were excluded from this review and the reason for exclusion

References	Reason for exclusion
Lori D et al. A Coloured Pen Needle Education System Improves Insulin Site Rotation Habits: Results of a Randomized Study. Diabetes Ther. 2020;11(12):2979-2991	Wrong population
Justin O et al. Nodular amyloidosis in a diabetic patient with frequent hypoglycemia: sequlae of repeatedly injecting insulin without site rotation. J Am Acad Dermatol. 2010;63(6):e113-114	Wrong study design
Inna V et al. A Randomized Controlled Trial to Assess the Impact of Proper Insulin Injection Technique Training on Glycemic Control. Diabetes Ther. 2017;8(6):1309-1318	Wrong population
Smith M et al. UK lipohypertrophy interventional study. Diabetes Res Clin Pract. 2017;126:248-253	Wrong population
Carina K et al. The roto track device improves rotation of insulin injections in T1DM: a proof-of-concept study. Diabetes. 2020;69(Supplement 1)	Wrong population
Selvadurai S et al. Impact of pharmacist insulin injection re-education on glycemic control among type II diabetic patients in primary health clinics. Saudi Pharm J. 2021;29(7):670-676	Wrong population

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BMJ Open

Efficacy of education on injection technique for patients diagnosed with diabetes with lipohypertrophy: a systematic review and meta-analysis

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Efficacy of education on injection technique for patients diagnosed with diabetes with lipohypertrophy: systematic review and meta-analysis

Masahiro Ichikawa[a]*, Tomoaki Akiyama[a], Yasushi Tsujimoto[b], Keisuke Anan[c], Tadashi Yamakawa[a], Yasuo Terauchi[d]

[a]Department of Endocrinology and Diabetes, Yokohama City University Medical Center, Yokohama, Japan

[b]Department of Nephrology and Dialysis, Kyoritsu Hospital, Kawanishi, Japan

[c]Department of Healthcare Epidemiology, Graduate School of Medicine and Public Health, Kyoto University, Kyoto, Japan

[d]Department of Endocrinology and Metabolism, Yokohama City University School of Medicine, Yokohama, Japan

***Corresponding Author**

Masahiro Ichikawa

Department of Endocrinology and Diabetes

Yokohama City University Medical Center

Urafunecho 4-57, Minami-ku

For peer review only

Yokohama 232-0024, Japan

Tel: +81-45-261-5656

Fax: +81-45-231-1846

E-mail: t196009c@yokohama-cu.ac.jp

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ABSTRACT

Objectives: This study aimed to investigate the efficacy of providing education on injection technique to patients with diabetes with lipohypertrophy (LH).

Design: We conducted a systematic review and meta-analysis.

Methods: We included patients with diabetes who use insulin and have LH, and excluded patients without LH. We performed a literature search on CENTRAL, MEDLINE, EMBASE, ICTRP, and ClinicalTrials.gov on November 2021 for randomised controlled trials (RCTs). We used the revised Cochrane Risk of Bias 2 tool to evaluate the risk of bias in each outcome in each study. We then pooled the data using a random-effects model and evaluated the certainty of evidence using the Grading of Recommendations Assessment, Development, and Evaluation approach.

Outcome measures: The primary endpoints were change in total daily dose (TDD) of insulin, change in HbA1c levels, and prevalence of hypoglycaemia.

Results: We screened 580 records and included three RCTs (637 participants) in the meta-analysis. Education on injection technique may slightly increase the change of TDD of insulin (3 studies, 637 participants: mean difference (MD) -6.26; 95% confidence interval (CI) -9.42, -3.10; $p < 0.001$; $I^2 = 38\%$; low certainty of evidence) and may have little to no effect on change in HbA1c but the evidence is very uncertain compared to that

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in the control group (three studies, 637 participants: MD -0.59; 95% CI -1.71, 0.54; p = 0.31; I^2 = 98%; very low certainty of evidence). Providing education about injection technique may have little-to-no effect on the prevalence of hypoglycaemia (three studies, 637 participants: risk ratio 0.44; 95% CI 0.06, 3.13; p = 0.41; I^2 = 90%; very low certainty of evidence).

Conclusions: The present meta-analysis suggests that injection technique education may result in a slight reduction in the TDD of insulin. However, the effect of education on HbA1c, hypoglycaemia, and cured LH is uncertain.

Protocol registration: DOI: dx.doi.org/10.17504/protocols.io.btiinkce

Keywords: injection technique, lipohypertrophy, diabetes mellitus

Strengths and limitations of this study

・We conducted a comprehensive search and summarised the evidence for the effects of injection technique education.

・We used a prospective registered protocol, followed the Cochrane Handbook and GRADE, selected a rigorous methodology, and adhered to the reporting guidelines.

・The limitation of this study was the fact that we did not have sufficient data to perform a subgroup analysis and could not evaluate the long-term effects of education on injection technique.

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INTRODUCTION

There has been a continuous increase in the number of diabetes patients worldwide, and diabetes has become a major public health problem, with 4.2 million adults estimated to die due to diabetes and its complications. [1] The number of diabetes patients is expected to increase to 700 million by 2045. [1] Blood glucose control is important to prevent the complications of diabetes, and insulin therapy plays a key role in regulating blood glucose level in patients. [2] Insulin therapy is necessary not only for type 1 diabetes but also for type 2 diabetes and is estimated to be used in 7.5% of type 2 diabetes patients. [3] Lipohypertrophy (LH), a side effect of long-term insulin injection, has been reported in 38% of diabetes patients. [4] LH is caused by the thickening of adipose tissue localised to the site of insulin injection. [5, 6]

Patients with LH face insulin variability because both insulin absorption and action becomes slow. [4, 7, 8] Moreover, the risk of unexpected hypoglycaemia has been noted in patients with LH. [7] Previous studies have suggested that providing proper education on injection techniques may reduce HbA1c levels and/or total daily dose (TDD) of insulin. [9-13] Injection techniques education (IT) includes rotating the injection sites properly, avoiding needle reuse, and avoiding injecting into the site of LH. [5] It is also recommended to use a 4-mm-long needle to avoid injection into the muscle. [5, 14]

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To the best of our knowledge, there has been no systematic investigation on the effects of providing education about injection technique on the TDD of insulin, HbA1c levels, and hypoglycaemia in diabetes patients with LH. Therefore, we aimed to perform a systematic review and meta-analysis to determine the efficacy of providing education on the injection technique to diabetes patients with LH.

METHODS

Ethics approval

As this was a systematic review, there were no ethical issues involved. We registered the protocol in protocols.io (DOI: dx.doi.org/10.17504/protocols.io.btiinkce). Patient consent was not necessary for the present study.

Eligibility criteria

We conducted this systematic review in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, [15] and the checklist is shown in the online supplemental table S1. We conducted a systematic review investigating the effects of IT on patients with type 1 or type 2 diabetes who use insulin and have LH, and included individual, cluster, and crossover randomised trials. We did

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not limit the inclusion to languages or countries. We included published and unpublished papers, conference abstracts, letters, and all other types of papers. We excluded observational studies and non-randomised controlled trials (RCTs). We did not exclude studies because of the observation period or year of publication.

We included patients with type 1 or type 2 diabetes who had been receiving insulin injections for at least 1 year, with LH, 18 years old and above, any sex, no HbA1c restriction, and no education about LH within the past 6 months. LH diagnosis was made by a physician, nurse, or investigators when it was clinically visible, palpable, or visible on ultrasound. [4] We accepted any number of insulin injections per day or needle size for insulin injections.

We excluded pregnant women, patients who wished to become pregnant, lactating women, and patients taking medications that may cause LH (anti-retroviral or corticosteroid therapy).

Information sources

Search strategy

We searched the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE using PubMed, and EMBASE using ProQuest Dialog on 11 November 2021.

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We also searched the World Health Organization International Clinical Trials Platform Search Portal (ICTRP) and ClinicalTrials.gov on 11 November 2021 for ongoing or unpublished trials. The search strategies are presented in the online supplemental Table S2. We checked the reference lists of the studies, including international guidelines, and identified the reference lists of eligible studies and articles citing eligible studies.

Selection process

Two independent reviewers (MI and TA) screened the titles and abstracts, followed by an assessment of eligibility based on the full texts. We contacted the original authors if the relevant data were missing. Disagreements between the two reviewers were resolved by discussion, and if this was not possible, a third reviewer (YT or KA) became an arbiter.

Data collection process

Two reviewers (MI and TA) independently extracted data from included studies using a standardised data collection form. Disagreements were resolved by discussion, and if this was not possible, a third reviewer (YT or KA) became an arbiter. We conducted up to two author queries by e-mail to obtain relevant data from the researchers.

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Data items

The primary endpoints were changes in insulin TDD and HbA1c levels, and the prevalence of hypoglycaemia. The change in insulin TDD was defined as the change in TDD from baseline, and the duration was the longest follow-up period after 3 months. The change in HbA1c levels was defined as the change in HbA1c from baseline, and the duration was the longest follow-up period after 3 months. Hypoglycaemia was defined as one or more hypoglycaemic symptoms (palpitations, fatigue, sweating, strong hunger, dizziness, tremors, etc.) and a blood glucose level below 60 mg/dL. [7] We accepted hypoglycaemia as defined by the original authors. Duration was the longest follow-up period after 3 months. The secondary endpoints were changes in the proportion of patients with cured LH and all adverse events. The definition of change in the proportion of patients with cured LH was the same as that of the original paper, and the duration was the longest follow-up period after 3 months. All adverse events were the same as those defined in the original study.

The data collecting form included information on the first author's name, year of publication, country, sample size, proportion of male participants, mean age of participants, number of participants with type 1 diabetes mellitus (T1DM)/type 2 diabetes mellitus (T2DM), estimated diabetes duration, mean duration of insulin treatment,

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intervention details, detection methods of LH, and the outcome measures mentioned above. We asked the authors of the original papers for unpublished or additional data and obtained research data for [13] but not for [12].

Study risk of bias assessment

Two reviewers (MI and TA) independently assessed the risk of bias using the Risk of Bias 2 (RoB2) tool. [16] The effect of interest is the intention-to-treat effect, which is the effect assigned to the interventions at baseline, regardless of whether or not the intervention was received as intended. Disagreements between the two reviewers were discussed, and if this was not possible, a third reviewer (YT or KA) became an arbiter.

Effect measures

We pooled the relative risk ratios and 95% confidence intervals (CIs) for the following binary variables: hypoglycaemia and proportion of patients with cured LH. We pooled the mean differences and 95% CIs for the following continuous variables: HbA1c and TDD of insulin.

We extracted the data on an intention-to-treat basis for all dichotomous data whenever possible. For continuous data, we did not impute missing data based on the

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recommendation of the Cochrane Handbook. [17] We performed a meta-analysis of the data available in the original study.

When the original studies reported only standard errors or p-values, we used Altman's method to calculate the standard deviation. [18] When these values were not available after enquiring the authors, we calculated the standard deviations using confidence intervals and t-values based on the methods in the Cochrane Handbook [17] or validated methods. [18] We analysed the validity of these methods using sensitivity analysis.

Statistical analyses

We used Review Manager software (RevMan 5.4; The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark) to conduct a meta-analysis using the DerSimonian and Laird random-effects method. We assessed statistical heterogeneity by visual inspection of the forest plots and calculation of I^2 statistic (I^2 values of 0-40%: might not be important; 30-60%: may represent moderate heterogeneity; 50-90%: may represent substantial heterogeneity; 75-100%, considerable heterogeneity). For the I^2 statistic, we performed the Cochrane Chi^2 test (Q test) and considered it statistically significant if the P value was less than 0.10.

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Certainty assessment

Based on the Cochrane Handbook, [17] we created a summary of findings (SoF) table for the results of changes in the insulin TDD, changes in HbA1c levels, and the prevalence of hypoglycaemia. A SoF table contained important information on the relative and absolute effect sizes of the interventions investigated, the amount of available evidence, and the certainty of available evidence. [17]

We assessed the quality of evidence for each outcome based on the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) method. [19] GRADE specifies four categories, namely high, moderate, low, and very low, which are applied to accumulating evidence. Evidence from randomised trials starts with high certainty of evidence and can be downgraded by five domains: risk of bias, imprecision, inconsistency, indirectness, and publication bias.

Difference between a protocol and a review

The duration of hypoglycemia was defined as the follow-up period, but as it was reported in terms of the number of patients who had hypoglycemia, the duration was defined as the longest follow-up after 3 months. We could not summarize all adverse events because there were few reports other than hypoglycaemia. To clarify the influence of effect

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modifiers on the results, subgroup analyses of the primary outcome were planned to be evaluated with the following factors when sufficient data were available: age (< 65 years vs. ≥ 65 years), baseline HbA1c level of < 8% vs. ≥ 8%, [20] sex, and IT protocol with or without initial insulin level reduction.

We also planned to conduct the following sensitivity analyses on the primary outcomes to assess whether the results of the review were robust to the decisions made during the review process: exclusion of studies with imputed statistics, and exclusion of studies with an overall high risk of bias. We could not perform a predetermined subgroup analysis and sensitivity analysis because of insufficient data.

We planned to assess the possibility of publication bias by visually inspecting the funnel plots and testing the Egger's test. If there were fewer than 10 trials or trials with similar sample sizes, then we decided not to perform the test. We could not perform the Egger test for reporting bias detection because there were fewer than 10 trials.

Patients and public involvement

As this was a systematic review, patients or the public were not involved in our research.

RESULTS

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Study selection

We performed the search in November 2021 and found 580 records; after duplicates were removed, 550 records were screened for titles and abstracts. Nine full-text records were assessed for eligibility. Six studies were excluded, and the reasons were as follows: five for including the wrong population and one for using the wrong design. Details of the excluded studies are shown in the online supplemental Table S3. Finally, we included three RCTs that met the eligibility criteria (Figure 1). [12, 13, 21]

Study characteristics

Table 1 presents the characteristics of the three included studies conducted in France, China, and Italy. In the French study, the number of individuals with T1DM/T2DM was 66/57, and in the Italian study, it was 0/318. In the Chinese study, there were fewer than seven patients with T1DM, but the exact number of patients with T1DM/T2DM could not be obtained despite inquiries. In the French study, all participants received information about optimal IT and LH at baseline for ethical reasons. In the Italian study, the proportion of cured LH was not reported.

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Table 1. Characteristics of the included studies.

Study	Methods	Subject characteristics	Intervention details	Reported outcome of interest
Campions 2017 [12]	Location: France Sample size: 123 Detection method of LH: visible and/or palpable LH determined by nurses	Age (years): 52.8 ± 15.1 (IT), 51.4 ± 16.4 (C) Male (%): 70.7 Number of T1DM/T2DM individuals: 66/57 Estimated diabetes duration (years): 19.8 ± 11.5 (IT), 16.3 ± 9.2 (C) Insulin treatment duration (years): 14.8 ± 12.8 (IT), 12.5 ± 10 (C)	IT: reducing insulin doses initially by 20%, instructing to rotate within injection sites, foregoing needle reuse, stopping injecting into the LH site and switching to 4-mm needles. C: standard care. All participants received information about optimal IT and LH at baseline.	Change in TDD of insulin from baseline Change in HbA1c levels from baseline Hypoglycaemia Change in the proportion of patients with cured LH All adverse events
Chen 2021 [13]	Location: China Sample size: 210 Detection method of LH: visible and/or palpable LH determined by nurses and ultrasound examination	Age (years): 59 ± 7 (IT), 60 ± 9 (C) Male (%): 46.7 Number of T1DM/T2DM individuals: N/A Estimated diabetes duration (years): 14.4 ± 5.8 (IT), 16.8 ± 7.6 (C) Insulin treatment duration (years): 6.9 ± 4.7 (IT), 7.1 ± 5.4 (C)	IT: reducing insulin doses initially by 20%, instructing to rotate within injection sites, foregoing needle reuse, stopping injecting into the LH site and switching to 4-mm needles. C: standard care	Change in TDD of insulin from baseline Change in HbA1c levels from baseline Hypoglycaemia Change in the proportion of patients with cured LH All adverse events

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Gentile 2021 [21]	Location: Italy Sample size: 318 Detection method of LH: inspection and palpation by healthcare professionals and ultrasound-based skin evaluations	Age (years): 61 ± 10 (IT), 63 ± 12 (C) Male (%): 44.0 Number of T1DM/T2DM individuals: 0/318 Estimated diabetes duration (years): 11.6 ± 9.8 (IT), 11.3 ± 5.7 (C) Insulin treatment duration (years): 6.5 ± 9.3 (IT), 6.7 ± 7.2 (C)	IT: reducing insulin doses initially by 20%, repeatedly instructing to rotate within injection sites, foregoing needle reuse and stopping injecting into the LH site. Change to 4 mm needles 6 months before randomization. Insulin titration. C: reducing insulin doses initially by 20%, at the first session instructing to rotate within injection sites, foregoing needle reuse and stopping injecting into the LH site. Change to 4 mm needles 6 months before randomization. Insulin titration.	Change in TDD of insulin from baseline Change in HbA1c levels from baseline Hypoglycaemia All adverse events

DM, Diabetes mellitus; IT, Injection technique education group; C, Control group; LH, lipohypertrophy; TDD, Total daily doses; N/A, Not available

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Risk of bias in the studies

Figure 2 shows the RoB2 result: In two of three RCTs, [12, 13] domain 2 (bias due to deviations from the intended intervention) and/or domain 3 (bias due to missing outcome data) were at high risk, and the overall risk of bias was high.

Outcome and certainty of evidence

Table 2 shows a summary of these findings. Figure 3 shows a forest plot. Compared with standard care, IT slightly increased the change in TDD of insulin (three studies, 637 participants: MD -6.26; 95% confidence interval (CI) -9.42, -3.10; p <0.001; I^2 = 38%; low certainty of evidence). The certainty of evidence was low because of serious risk of bias and imprecision. Compared with standard care, IT may increase little to no effect on the change in HbA1c levels but the evidence is very uncertain (three studies, 637 participants: MD -0.59; 95% CI -1.71, 0.54; p = 0.31; I^2 = 98%; very low certainty of evidence). The certainty of evidence was very low because of serious risk of bias, imprecision, and inconsistency.

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Table 2. Summary of findings comparing indicated outcomes between injection technique education and standard care.

Patient or population: Patients with type 1 diabetes or type 2 diabetes who use insulin and have LH

Setting: Outpatient

Intervention: Injection techniques education

Comparison: Standard care

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Quality of evidence (GRADE)
	Assumed risk-C	Corresponding risk-IT			
Change in TDD of insulin	Mean change in the TDD of insulin in the C group was -3.12 to -0.9.	Mean change in the TDD of insulin in the IT group was 6.26 lower(9.42 lower to 3.10 lower)		637 (3)	Low[a,b]
Change in HbA1c level	Mean change in HbA1c level in the C group was -0.55 to 0.1.	Mean change in HbA1c level in the IT group was 0.59 lower(1.71 lower to 0.54 higher)		637 (3)	Very low[a,b,c]
Prevalence of hypoglycaemia	162/319 (50.8%)	21/318 (6.60%)	RR 0.44 (0.06 to 3.13)	637 (3)	Very low[a,c,d]

* Corresponding risk (and 95% CI) are based on assumed risks in the control group and relative risks (and 95% CI) of the intervention. [a] Due to serious risk of bias
[b] Due to serious imprecision. The sample size did not meet the optimal size criterion.
[c] Due to serious inconsistency
[d] Due to very serious imprecision. The sample size did not meet the optimal size criterion, and the 95% CI was wide.LH, lipohypertrophy; C, Control; IT, Injection techniques education; TDD, total daily doses; HbA1c, haemoglobin A1c; RR, risk ratio; CI, confidence interval. GRADE Working Group grades of evidence: High quality: we are very confident that the true effect lies close to that of the estimate of the effect. Moderate quality: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low quality: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect. Very low quality: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

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The evidence is very uncertain about the effect of insulin injection technique on the prevalence of hypoglycaemia (three studies, 637 participants: risk ratio 0.44; 95% CI 0.06, 3.13; p = 0.41; I^2 = 90%; very low certainty of evidence). The very low certainty of evidence was due to the serious risk of bias and inconsistency and the very serious imprecision. The evidence is very uncertain about the prevalence of cured LH compared to the control group (two studies, 319 participants: RR 1.19; 95% CI 0.73, 1.92; p = 0.49; I^2 = 0%; very low certainty of evidence). The very low certainty of evidence was due to the serious risk of bias and very serious imprecision.

DISCUSSION

The present study showed that IT may result in a slight reduction in the TDD of insulin. The evidence is very uncertain about the effect of insulin injection technique on the change in HbA1c levels, the prevalence of hypoglycaemia and the prevalence of cured LH compared to those in the control group. To the best of our knowledge, this is the first systematic review and meta-analysis to study the effects of insulin IT in comparison with standard care for diabetes patients with LH.

The results of the present study were partly consistent with those of previous observational studies, which found that IT was associated with changes in TDD of insulin and HbA1c levels. [9, 10] These previous observational studies reported that IT decreased the TDD of insulin and HbA1c levels in patients with diabetes upon insulin injection. Our meta-analysis of randomised trials showed similar results of TDD of insulin. Meanwhile, the results of the present study did not show a decrease in HbA1c levels, but this might be due to the fact that the three studies included were initially conducted with up to 20% reduction in TDD of insulin to prevent hypoglycaemia.

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Excessive exogenous insulin can be harmful, and it is important to reduce the TDD of insulin to prevent adverse events. Excessive exogenous insulin administration has been reported to be associated with adverse events such as hypoglycaemia, weight gain, and increased cardiovascular events and mortality. [22, 23] A meta-analysis in type 1 diabetes patients integrated aerobic-only intervention studies in adults, and multiple types of exercise interventions, including aerobic and anaerobic, in 12- to 18-year-olds. The results showed a reduction of 0.21 U/kg in TDD of insulin. [24] The present study showed a decrease in insulin TDD of 6.26 U/day with improved pharmacokinetics/pharmacodynamics by injection technique education. Therefore, although the effects obtained by exercise interventions, which have different mechanism, showed greater results, injection technique education may be effective in avoiding excessive doses of insulin.

The certainty of evidence in the present study was low or very low. In one included study, [13] a portion of the control group was shifted to the ideal injection method, which contributed to the effect of high risk of bias due to deviation from the intended intervention. In the other two studies, some of the control groups was shifted to the ideal injection method. [12, 21] Despite the fact that some of the control group shifted to the ideal injection method, the present study showed a slight reduction in the TDD of insulin, which underestimates the effect. In fact, the effect may be even greater. In two of three studies, [12, 13] the bias due to missing outcome data for the continuous variable outcome was high because the missing value was greater than 5%. The bias due to missing outcome data for the binary variable outcome was low or high for the frequency of outcome and number of missing outcomes. The registration of the prevalence of hypoglycaemia in clinicaltrials.gov mentions only a timeframe of 6 months, but because the prevalence of

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hypoglycaemia was counted at 3 and 6 months, we considered the bias in the selection of the [12] as high. The other reason for lowering the GRADE is imprecision, because only three studies were included, and the sample size was small. In the future, more studies with sufficient sample sizes are needed.

The strengths of this study are that we conducted a comprehensive search and summarised the evidence for the effects of IT, used a prospective registered protocol, followed the Cochrane Handbook and GRADE, selected a rigorous methodology, and adhered to reporting guidelines.

However, this study had several limitations. First, as described above, we could not have enough data to conduct subgroup analyses and assess publication bias.It was not plausible for publication bias to affect our findings because our comprehensive search did not show any ongoing or unpublished studies. Second, the generalisability of our findings are limited. For example, the included studies tended to recruit middle-aged, obese, and patients with type 2 diabetes patients, and measured the outcomes at 6 months. However, a crossover study limited to type 1 diabetes individuals has shown that injections administered at sites away from the LH improved insulin absorption and reduce glycaemic variability compared to injections administered at the LH. [8] Previous studies have also shown the efficacy of optimised injection technique on TDD of insulin, although baseline characteristics varied, suggesting that it may be effective in the aforementioned populations. [9, 10] Further studies evaluating different populations and long-term outcomes are required. Third, we could not perform a subgroup analysis of types of DM which was requested during peer-review due to the limited data. The prevalence of LH was higher in type 2 diabetes than in type 1 diabetes, thus the types of DM may be an effect modifier of IT. [4] Future studies that use individual participant data

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can assess the effect modification. Fourth, we could not evaluate the effects beyond 6 months in this study. The two studies, i.e., ISTERP-1 [21] and ISTERP-2,[25] had a 12-month follow-up duration, and although the ISTERP-2 study did not meet the eligibility criteria of this study, their findings suggest that repeating the education on injection technique has lasting effects.[21, 25]

CONCLUSION

The present meta-analysis suggests that providing education on injection technique to diabetes patients with LH may slightly reduce the TDD of insulin. Clinicians may consider providing IT to patients with diabetes with LH, but they also need to understand the uncertainty of the evidence. In the future, we hope that RCTs with less bias, especially for the deviation of the intended protocol and with a rigorous protocol, will be conducted to accurately evaluate the effects.

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COMPETING INTERESTS:

All authors have no conflicts of interest to declare.

AUTHORS' CONTRIBUTIONS

MI conducted the search and analysed the data. MI and TA conducted screening of titles and abstracts, extracted data, and evaluated the risk of bias. If the disagreement could not

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be resolved, then YT or KA became an arbiter. MI, YT, and KA drafted the manuscript. YT, KA, TY, and YT revised the manuscript. All authors approved the manuscript version submitted for publication and agreed to be responsible for all aspects of this work.

ETHICAL APPROVAL: Not required.

PATIENT CONSENT FORM

Not required

DATA SHARING STATEMENT The dataset is available from the corresponding author upon request.

References

1 *Federation ID. IDF diabetes atlas*. 9th ed. http://www.diabetesatlas.org/ (Accessed 11 Jun 2021).

2 Brown A, Guess N, Dornhorst A, et al. Insulin-associated weight gain in obese type 2 diabetes mellitus patients: What can be done? *Diabetes Obes Metab* 2017;**19**:1655–68.

3 Basu S, Yudkin JS, Kehlenbrink S, et al. Estimation of global insulin use for type 2 diabetes, 2018–30: A microsimulation analysis. *Lancet Diabetes Endocrinol* 2019;**7**:25–33.

4 Deng N, Zhang X, Zhao F, et al. Prevalence of lipohypertrophy in insulin-treated diabetes patients: A systematic review and meta-analysis. *J Diabetes Investig*

For peer review only

2017;**9**:536–43.

5 Frid AH, Kreugel G, Grassi G, et al. New insulin delivery recommendations. *Mayo Clin Proc* 2016;**91**:1231–55.

6 Gentile S, Guarino G, Corte TD, et al. Insulin-induced skin lipohypertrophy in type 2 diabetes: A multicenter regional survey in Southern Italy. *Diabetes Ther* 2020;**11**:2001–17.

7 Blanco M, Hernández MT, Strauss KW, et al. Prevalence and risk factors of lipohypertrophy in insulin-injecting patients with diabetes. *Diabetes Metab* 2013;**39**:445–53.

8 Famulla S, Hövelmann U, Fischer A, et al. Insulin injection into lipohypertrophic tissue: Blunted and more variable insulin absorption and action and impaired postprandial glucose control. *Diabetes Care* 2016;**39**:1486–92.

9 Grassi G, Scuntero P, Trepiccioni R, et al. Optimizing insulin injection technique and its effect on blood glucose control. *J Clin Transl Endocrinol* 2014;**1**:145–50.

10 Smith M, Clapham L, Strauss K. UK lipohypertrophy interventional study. *Diabetes Res Clin Pract* 2017;**126**:248–53.

11 Misnikova IV, Gubkina VA, Lakeeva TS, et al. A randomized controlled trial to assess the impact of proper insulin injection technique training on glycemic control. *Diabetes Ther* 2017;**8**:1309–18.

12 Campinos C, Le Floch JP, Petit C, et al. An effective intervention for diabetic lipohypertrophy: Results of a randomized, controlled, prospective multicenter study in France. *Diabetes Technol Ther* 2017;**19**:623–32.

13 Chen L, Xing Q, Li J, et al. Injection technique education in patients with diabetes injecting insulin into areas of lipohypertrophy: A randomized controlled trial.

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Diabetes Ther 2021. doi: 10.1007/s13300-021-01013-1 [Published online first: 2021/02/12].

14 Frid AH, Hirsch LJ, Menchior AR, et al. Worldwide injection technique questionnaire study: Injecting complications and the role of the professional. *Mayo Clin Proc* 2016;**91**:1224–30.

15 Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71.

16 Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;366:l4898.

17 Higgins JPT TJ. In: Chandler J, Cumpston M, Li T, Page MJ, Welch VA, (editors). http://www.training.cochrane.org/handbook. *Cochrane Handbook for Systematic Reviews of Interventions version 6.0 Cochrane* 2019.

18 Furukawa TA, Barbui C, Cipriani A, et al. Imputing missing standard deviations in meta-analyses can provide accurate results. *J Clin Epidemiol* 2006;**59**:7–10.

19 Guyatt G, Oxman AD, Akl EA, et al [GRADE guidelines]. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol* 2011;**64**:383–94.

20 Duke SA, Colagiuri S, Colagiuri R. Individual patient education for people with type 2 diabetes mellitus. *Cochrane Database Syst Rev* 2009;(1):CD005268.

21 Gentile S, Guarino G, Della Corte T, et al. Role of structured education in reducing Lypodistrophy and its metabolic complications in insulin-treated people with Type 2 diabetes: A randomized multicenter case-control study. *Diabetes Ther* 2021;**12**:1379–98.

22 McCall AL. Insulin therapy and hypoglycemia. *Endocrinol Metab Clin North Am*

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2012;**41**:57–87.

23 Herman ME, O'Keefe JH, Bell DSH, et al. Insulin therapy increases cardiovascular risk in type 2 diabetes. *Prog Cardiovasc Dis* 2017;**60**:422–34.

24 Yardley JE, Hay J, Abou-Setta AM, et al. A systematic review and meta-analysis of exercise interventions in adults with type 1 diabetes. *Diabetes Res Clin Pract* 2014;**106**:393–400.

25 Gentile S, Guarino G, Della Corte T, et al. The Durability of an Intensive, Structured Education-Based Rehabilitation Protocol for Best Insulin Injection Practice: The ISTERP-2 Study. *Diabetes Ther* 2021;12(9):2557–69.

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Figure Legends

Figure 1. PRISMA Flow Diagram.

Figure 2. Risk of bias evaluation for each outcome. The risk of bias was assessed using the Cochrane risk of bias tool for randomised trials. TDD, total daily dose; LH, lipohypertrophy

Figure 3. Forest plots of primary and secondary outcomes

A. Change in total daily doses of insulin

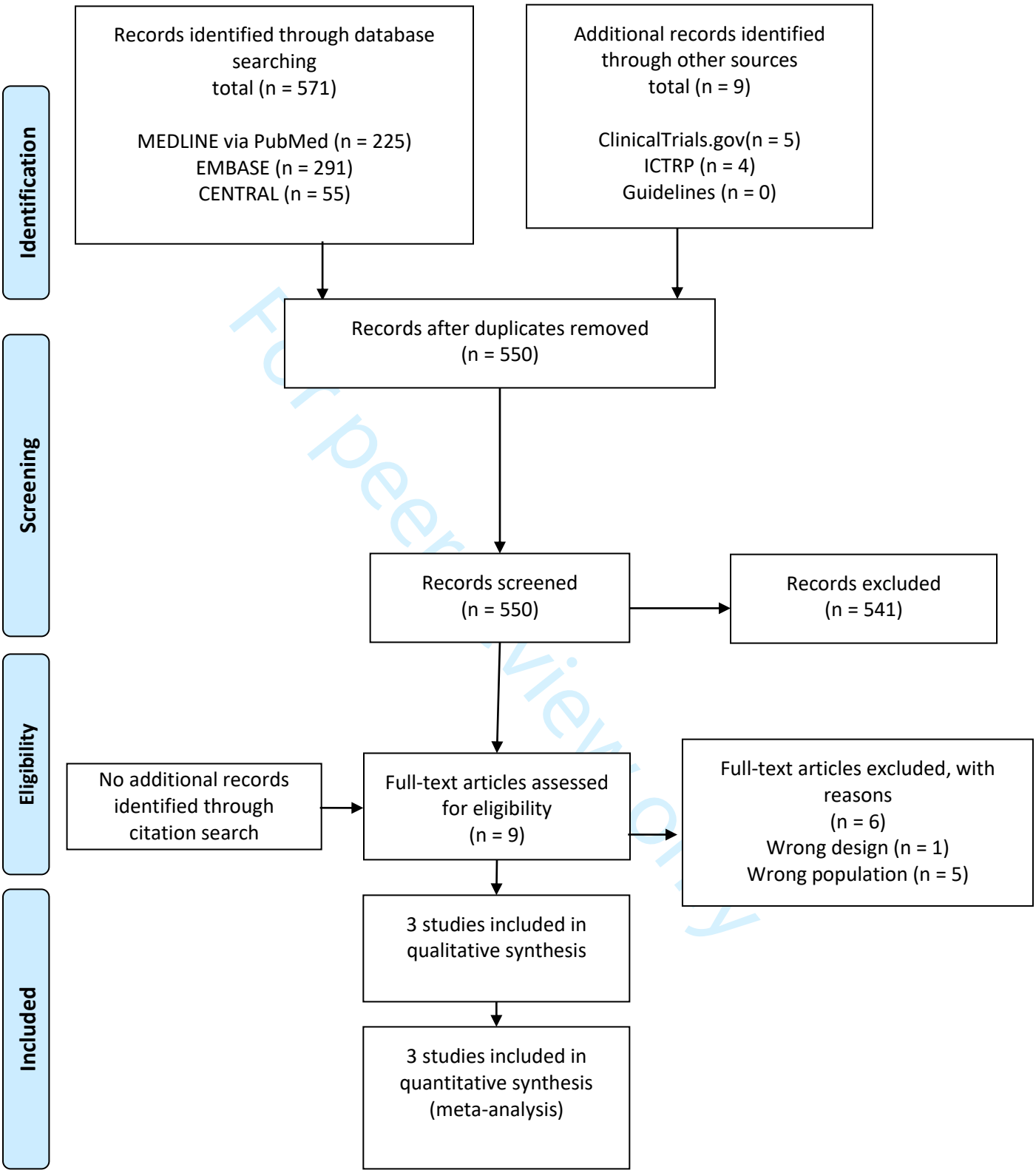
B. Change in HbA1c levels

C. Prevalence of hypoglycaemia

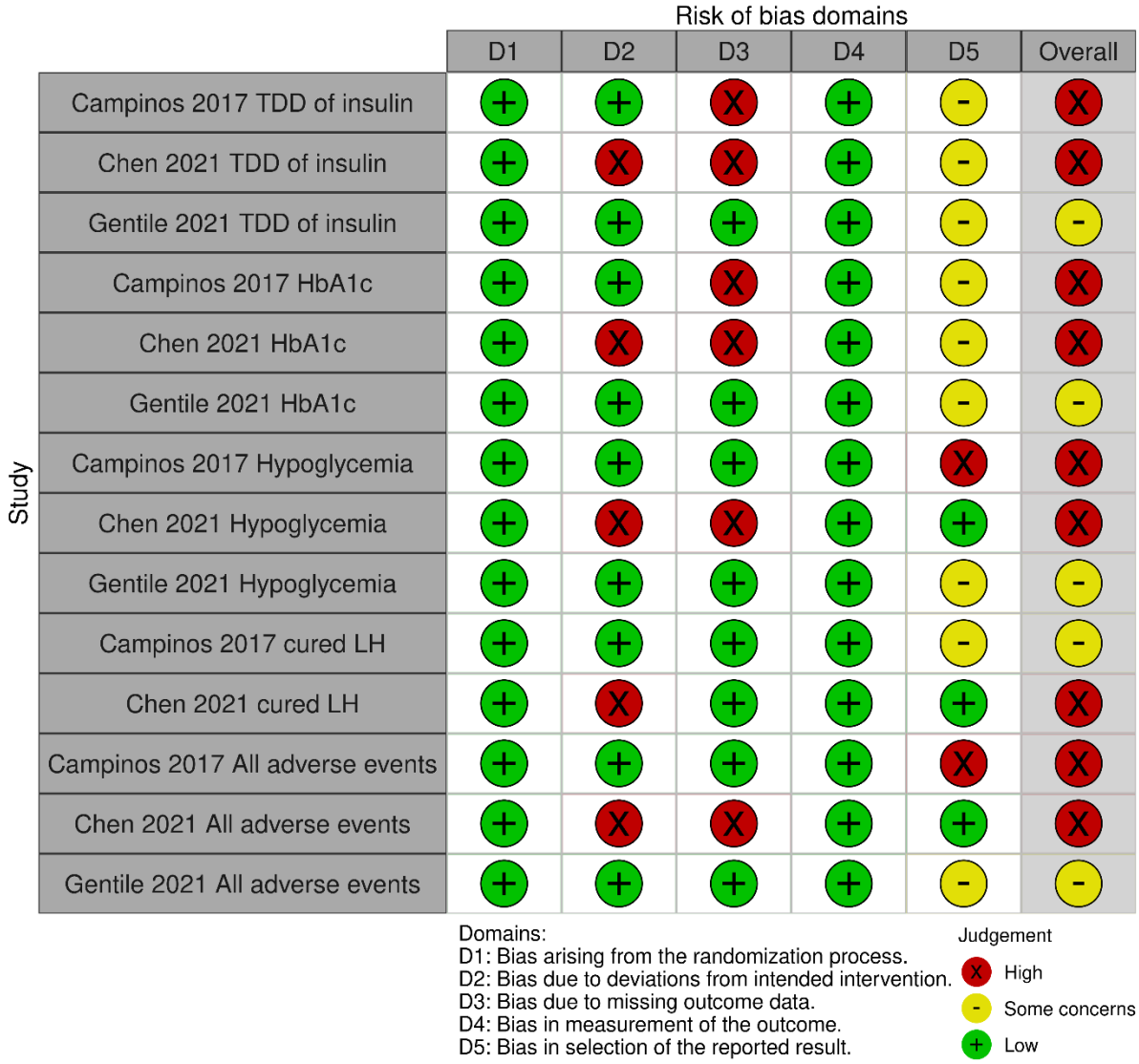
D. Change in the proportion of patients with cured LH

LH, lipohypertrophy

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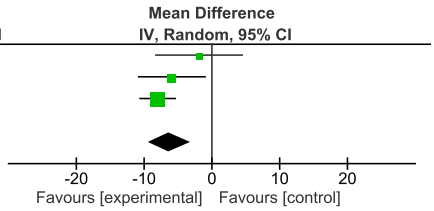
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Risk of bias domains
D1
D2
D3
D4
D5
Overall
Study
Campinos 2017 TDD of insulin
Chen 2021 TDD of insulin
Gentile 2021 TDD of insulin
Campinos 2017 HbA1c
Chen 2021 HbA1c
Gentile 2021 HbA1c
Campinos 2017 Hypoglycemia
Chen 2021 Hypoglycemia
Gentile 2021 Hypoglycemia
Campinos 2017 cured LH
Chen 2021 cured LH
Campinos 2017 All adverse events
Chen 2021 All adverse events
Gentile 2021 All adverse events
Domains:
D1: Bias arising from the randomization process.
D2: Bias due to deviations from intended intervention.
D3: Bias due to missing outcome data.
D4: Bias in measurement of the outcome.
D5: Bias in selection of the reported result.
Judgement
High
Some concerns
Low

A.

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Campinos 2017	-5.02	16.9065	53	-3.12	16.9902	56	19.0%	-1.90 [-8.27, 4.47]
Chen 2021	-6.8	19.1	106	-0.9	17	104	27.8%	-5.90 [-10.79, -1.01]
Gentile 2021	46	9	159	54	14	159	53.2%	-8.00 [-10.59, -5.41]
Total (95% CI)			**318**			**319**	**100.0%**	**-6.26 [-9.42, -3.10]**

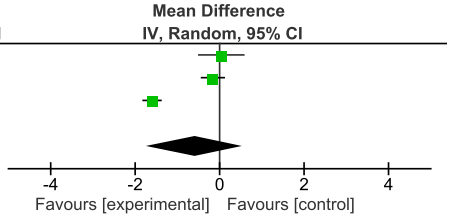
Heterogeneity: Tau² = 3.14; Chi² = 3.23, df = 2 (P = 0.20); I² = 38%
Test for overall effect: Z = 3.88 (P = 0.0001)



B.

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
Campinos 2017	-0.5	1.41	53	-0.54	1.42	56	32.0%	0.04 [-0.49, 0.57]
Chen 2021	-0.71	0.99	106	-0.55	0.98	104	33.9%	-0.16 [-0.43, 0.11]
Gentile 2021	6.6	0.9	159	8.2	1	159	34.1%	-1.60 [-1.81, -1.39]
Total (95% CI)			**318**			**319**	**100.0%**	**-0.59 [-1.71, 0.54]**

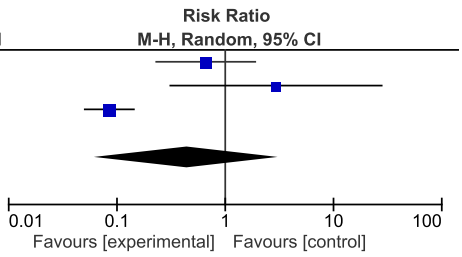
Heterogeneity: Tau² = 0.96; Chi² = 84.21, df = 2 (P < 0.00001); I² = 98%
Test for overall effect: Z = 1.02 (P = 0.31)



C.

Study or Subgroup	Experimental Events	Total	Control Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
Campinos 2017	5	53	8	56	35.5%	0.66 [0.23, 1.89]
Chen 2021	3	106	1	104	26.0%	2.94 [0.31, 27.84]
Gentile 2021	13	159	153	159	38.5%	0.08 [0.05, 0.14]
Total (95% CI)		**318**		**319**	**100.0%**	**0.44 [0.06, 3.13]**
Total events	21		162			

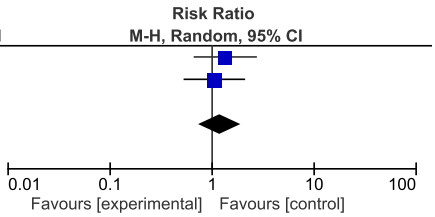
Heterogeneity: Tau² = 2.52; Chi² = 19.11, df = 2 (P < 0.0001); I² = 90%
Test for overall effect: Z = 0.82 (P = 0.41)



D.

Study or Subgroup	Experimental Events	Total	Control Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
Campinos 2017	14	53	11	56	48.7%	1.34 [0.67, 2.69]
Chen 2021	15	106	14	104	51.3%	1.05 [0.53, 2.07]
Total (95% CI)		**159**		**160**	**100.0%**	**1.19 [0.73, 1.92]**
Total events	29		25			

Heterogeneity: Tau² = 0.00; Chi² = 0.25, df = 1 (P = 0.62); I² = 0%
Test for overall effect: Z = 0.69 (P = 0.49)



Supplementary Tables:
Table S1 PRISMA checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6, 7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table S2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8, 9
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	9, 10
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	10, 11
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	6, 7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary	11, 12

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Section and Topic	Item #	Checklist item	Location where item is reported
		statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	12
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	11, 12
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	13
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	13
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	13
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	12
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	14
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	14
Study characteristics	17	Cite each included study and present its characteristics.	14, 15, 16
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	18, Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	18, 20
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	14, 15, 16, 18
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	18, 20, Figure 3
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	13
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	13
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 2
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	20
	23b	Discuss any limitations of the evidence included in the review.	22

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Section and Topic	Item #	Checklist item	Location where item is reported
	23c	Discuss any limitations of the review processes used.	22
	23d	Discuss implications of the results for practice, policy, and future research.	23
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	4, 6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	4, 6
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	13
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	23
Competing interests	26	Declare any competing interests of review authors.	23
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	24

Table S2 Search strategy

CENTRAL search strategy

#1 [mh Amyloidosis]
#2 lipohypertrophy:ti,ab
#3 "insulin-derived amyloidosis":ti,ab
#4 lipos:ti,ab
#5 "subcutaneous induration":ti,ab
#6 "subcutaneous nodules":ti,ab
#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8 [mh "Injections, Subcutaneous"]
#9 [mh "Patient Education as Topic"]
#10 injection*:ti,ab
#11 education*:ti,ab
#12 (reus*:ti,ab AND needle*:ti,ab)
#13 #8 OR #9 OR #10 OR #11 OR #12
#14 #7 AND #13

MEDLINE (via PubMed) search strategy

#1 "Amyloidosis"[Mesh]
#2 "lipohypertrophy"[tiab]
#3 "insulin-derived amyloidosis"[tiab]
#4 "lipos"[tiab]
#5 "subcutaneous induration"[tiab]
#6 "subcutaneous nodules"[tiab]
#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#8 "Injections, Subcutaneous"[Mesh]

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#9 "Patient Education as Topic"[Mesh]
#10 injection*[tiab]
#11 education*[tiab]
#12 reus*[tiab] AND needle*[tiab]
#13 #8 OR #9 OR #10 OR #11 OR #12
#14 #7 AND #13
#15 randomized controlled trial[pt]
#16 controlled clinical trial[pt]
#17 randomized[tiab]
#18 placebo[tiab]
#19 drug therapy[sh]
#20 randomly[tiab]
#21 trial[tiab]
#22 groups[tiab]
#23 #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22
#24 animals[mh] NOT humans[mh]
#25 #23 NOT #24
#26 #14 AND #25

EMBASE (via ProQuest Dialog) search strategy
#1 emb(amyloidosis) OR ti(lipohypertrophy) OR ab(lipohypertrophy) OR ti(insulin-derived amyloidosis) OR ab(insulin-derived amyloidosis) OR ti(lipos) OR ab(lipos) OR ti(subcutaneous induration) OR ab(subcutaneous induration) OR ti(subcutaneous nodules) OR ab(subcutaneous nodules)
#2 exact(subcutaneous drug administration) OR exact(patient education) OR ti(injection*) OR ab(injection*) OR ti(education*) OR ab(education*) OR (ti(reus*) OR ab(reus*)) AND (ti(needle*) OR ab(needle*))
#3 (ab(random*) OR ti(random*)) OR (ab(clinical NEAR/1 trial*) OR ti(clinical NEAR/1 trial*)) OR (EMB.EXACT("health care quality"))
#4 #1 AND #2 AND #3

ICTRP search strategy
Participant: lipohypertrophy OR insulin-derived amyloidosis OR subcutaneous induration OR subcutaneous nodules
Intervention: injection OR education OR (reus AND needle)

ClinicalTrials.gov search strategy
Condition or disease: lipohypertrophy OR insulin-derived amyloidosis OR subcutaneous induration OR subcutaneous nodules
Intervention: injection OR education OR (reus AND needle)

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Table S3. The list of studies that were excluded from this review and the reason for exclusion

References	Reason for exclusion
Lori D et al. A Coloured Pen Needle Education System Improves Insulin Site Rotation Habits: Results of a Randomized Study. Diabetes Ther. 2020;11(12):2979-2991	Wrong population
Justin O et al. Nodular amyloidosis in a diabetic patient with frequent hypoglycemia: sequlae of repeatedly injecting insulin without site rotation. J Am Acad Dermatol. 2010;63(6):e113-114	Wrong study design
Inna V et al. A Randomized Controlled Trial to Assess the Impact of Proper Insulin Injection Technique Training on Glycemic Control. Diabetes Ther. 2017;8(6):1309-1318	Wrong population
Smith M et al. UK lipohypertrophy interventional study. Diabetes Res Clin Pract. 2017;126:248-253	Wrong population
Carina K et al. The roto track device improves rotation of insulin injections in T1DM: a proof-of-concept study. Diabetes. 2020;69(Supplement 1)	Wrong population
Selvadurai S et al. Impact of pharmacist insulin injection re-education on glycemic control among type II diabetic patients in primary health clinics. Saudi Pharm J. 2021;29(7):670-676	Wrong population

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